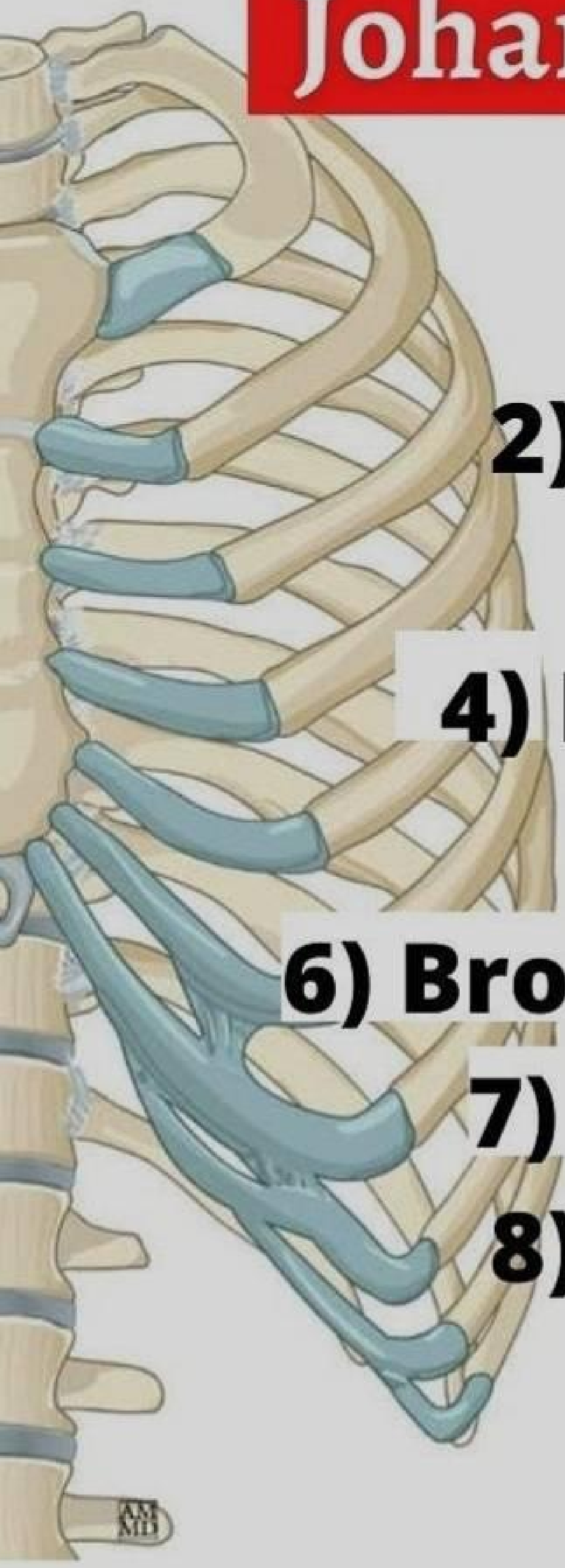
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PART-1

1) Parietal Pleura - 2

2) Pulmonary Ligament

3) Recess of Pleura

4) Lungs (Fissure & lobes)

5) Root of lungs

6) Bronchopulmonary Segments

7) Superior Mediastinum

8) Inferior Mediastinum

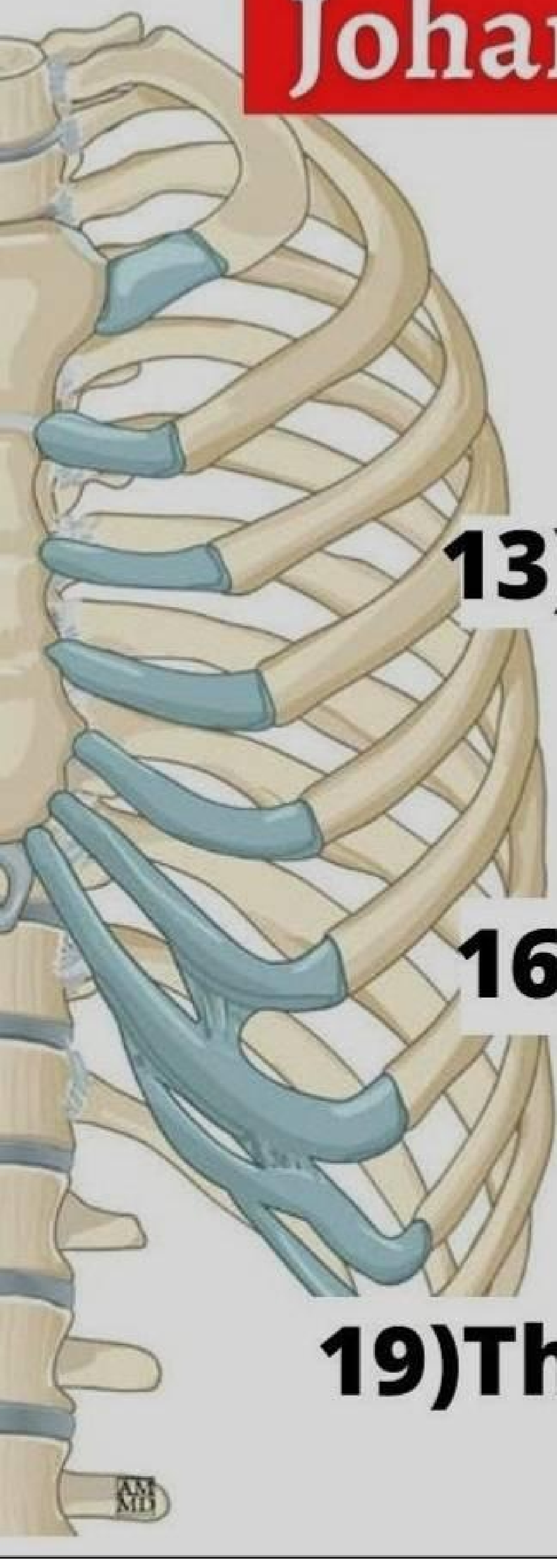
9) Pericardium

10) Right Atrium

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THORAX IMP TOPICS

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PART-2

11) Coronary Artery

12) Coronary Sinus

13) Nerve supply of Heart

14) Azygos Vein

15) Thoracic Duct

16) Thorax Wall & Spaces

17) Thoracic Nerve

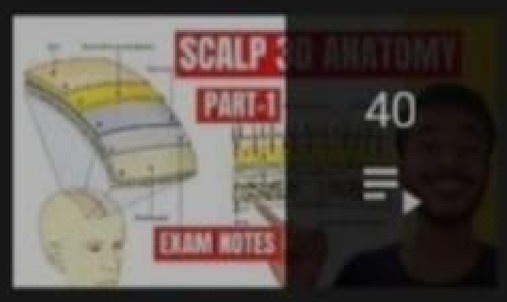
18) Thoracic Inlet

19)Thoracic Outlet|Diaphragm

20) Arch of Aorta

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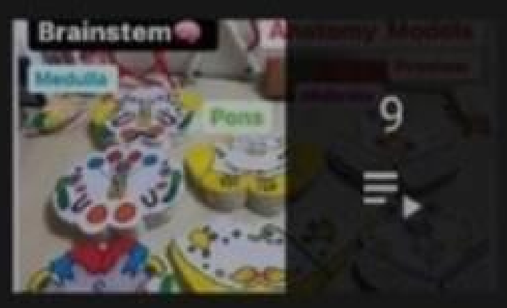
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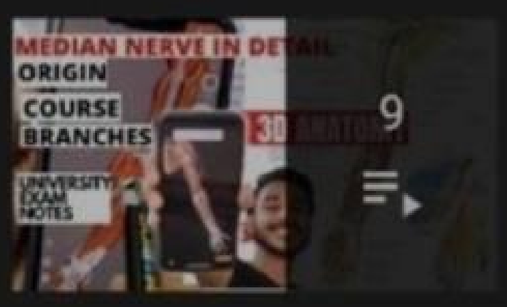
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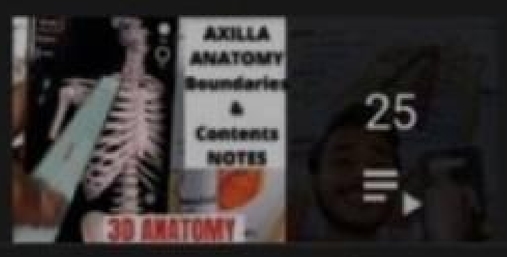
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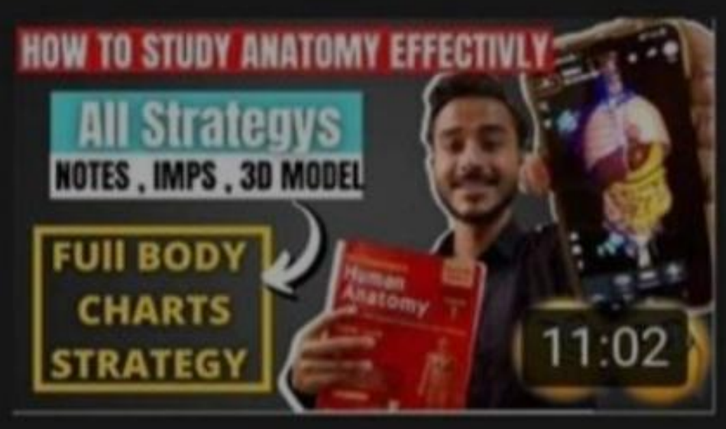
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PLEURA...

↳ Pleura is a serous membrane — lined by mesothelium
↳ 2 Pleural sac, 1 on either side of the mediastinum.
↳ Outer layer is parietal pleura
↳ Inner layer is visceral or pulmonary pleura.
↳ 2 pleura continue with each other around hilum of lungs blw pleural cavity.

Comparison of visceral & parietal pleura.

	Viseral	Parietal
Development	Splanopleuric Mesoderm	Somatopleuric mesoderm.
Position	Lines surface of lung including the tissues.	Line thoracic wall, media-stinum & diaphragm.
Nerve supply	Sympathetic nerves from T_2-T_5 ganglia Parasympathetic from vagus nerve.	Thoracic nerves & Phrenic nerves.
Sensitivity.	Insensitive to pain	sensitive to pain
Blood supply.	Bronchial vessels.	Intercostal & pericardiaco-phrenic vessels.
Lymphatic drainage TB	Tracheobronchial LN.	Intercoastal LN.

Visceral pleura :— Cover surface & fissures of lungs except hilium where it continue with parital pleura.

↳ Firmly adherent to the lungs can not be seperated.

Parietal pleura :— Thicker than viseral pleura. Sub divided into 4 parts. — Costal
Diaphragmatic
Mediastinal
Cervical.

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Surface marking of Pleura :— (Parietal)

↳ Cervical pleura — Represent by curved line.
↳ About 2.5 cm above clavicle.

↳ Ant. margin of the costomediastinal —

Right side — Extend — sternoclavicular joint to sternum angle.
vertically downward to xiphisternal point.

Left side — Follow the same course up to 4th costocartilage level.
↳ Then arch down to 6th costocartilage.

↳ Post margin — Pass 2 cm lat. to 12 thoracic spine
↓ To
2cm lat. to 7th cervicle spine.

↳ Inf. margin —
or
Costodiaphragmatic — cross 8th rib in mid clavicular line.
10th rib in mid axillary line.

Surface making of Pleura :— (Visceral)

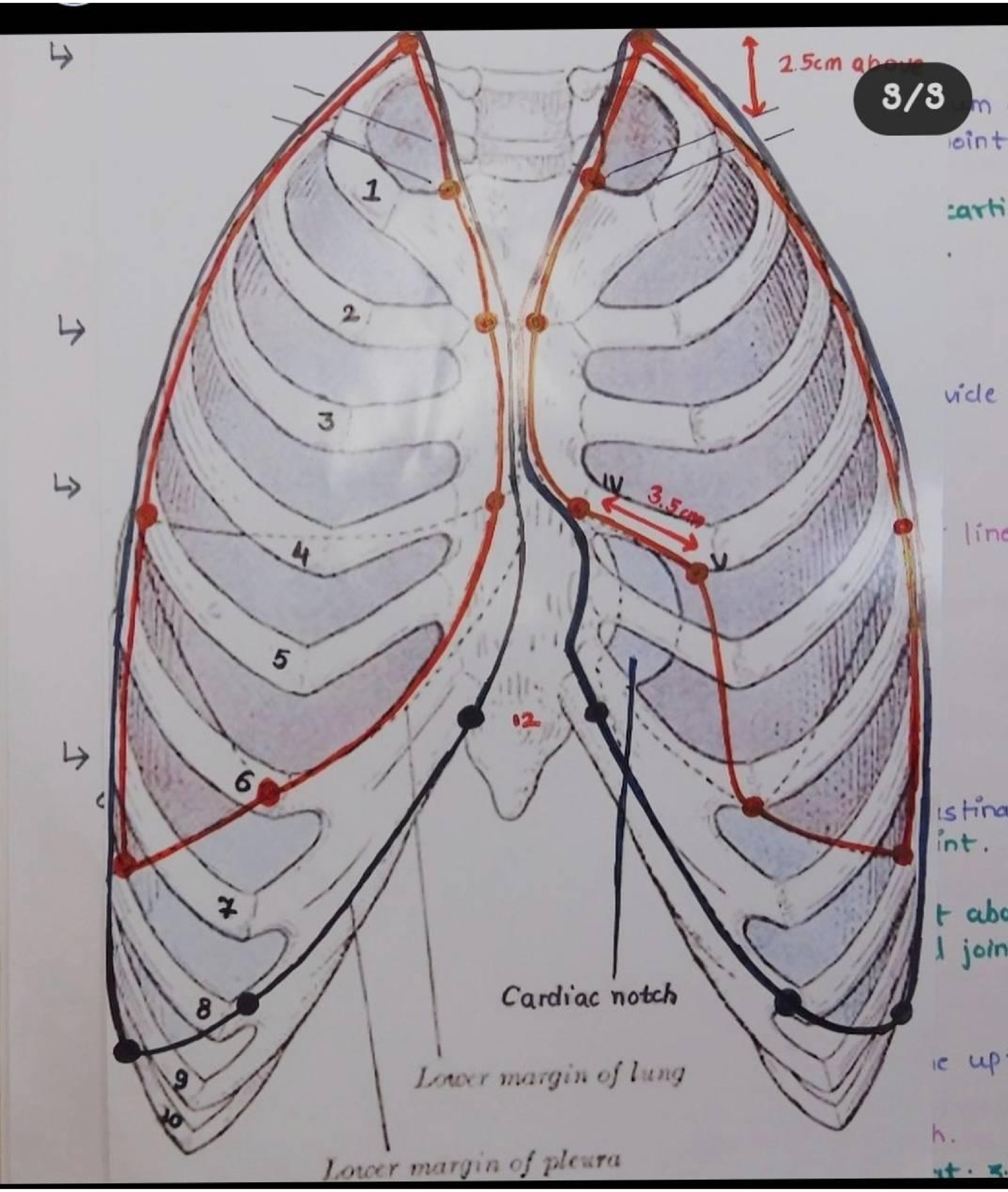
↳ Apex — 2.5 cm above medial 2/3 clavicle.
o— Ant. border — Rt side of ~~Rt~~ visceral pleura.
— Correspound closely with coasto mediastinal line.
— obtain by joining — sterno clavicular joint.
sternal angle.
Medial plane just above costa
to xiphisternal joint.

— Lt side of ~~Lt~~ visceral pleura.
— Correspound to coastomediastinal line up to level of 4th costo cartilage.
— Lower part represent a cardiac notch.
— From 4th coastocartilage it pass — lat. 3.5cm from sternal margin.
— Then curve down medially to reach 6th coasto cartilage.

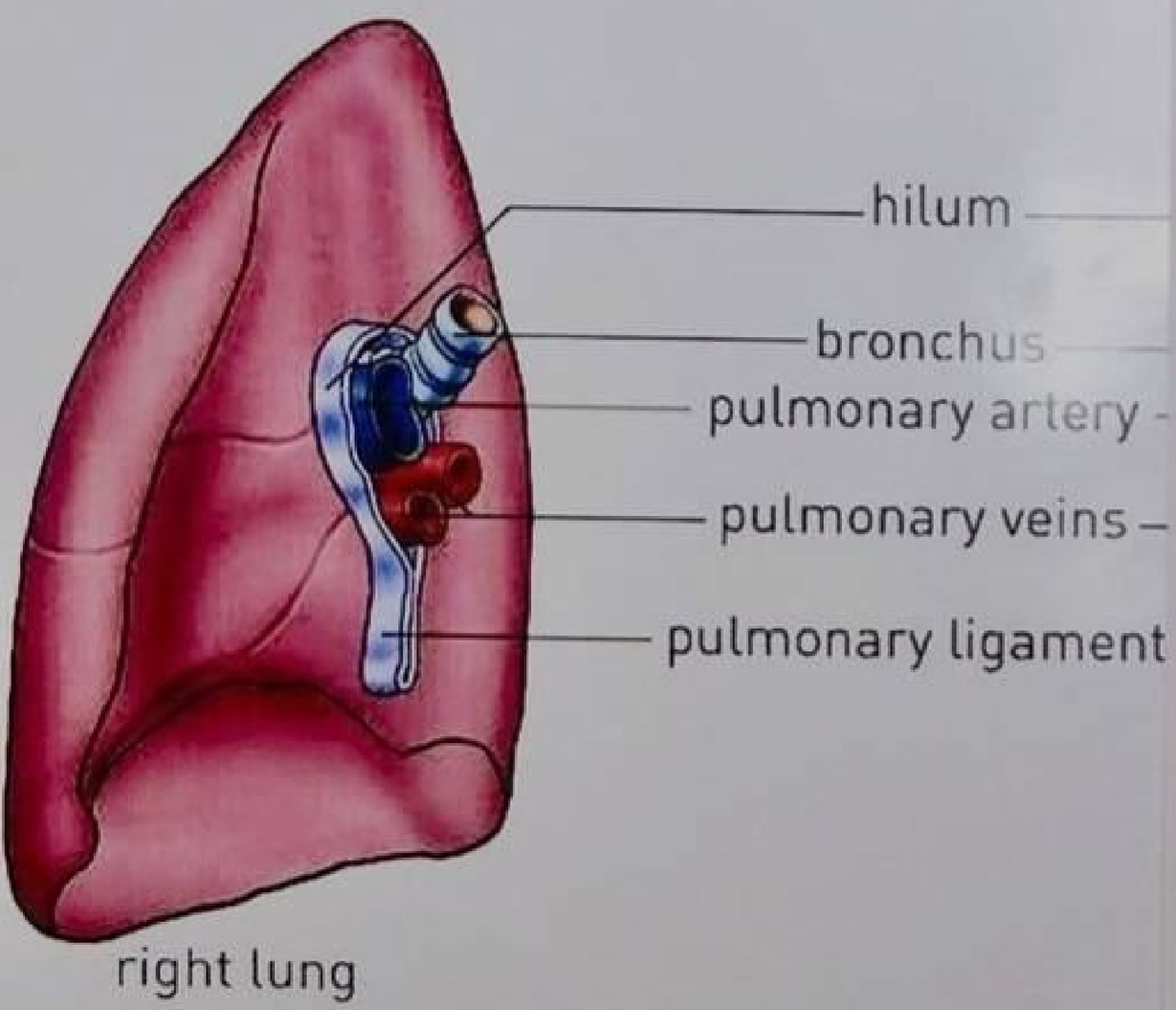
Area of cardiac notch is dull on percussion ~~cld~~
area of superficial cardiac dullness.

o— Lower border — Cross 6 rib in midclavicular line, 8th rib in mid axillary line.

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right lung

Pulmonary Ligament....

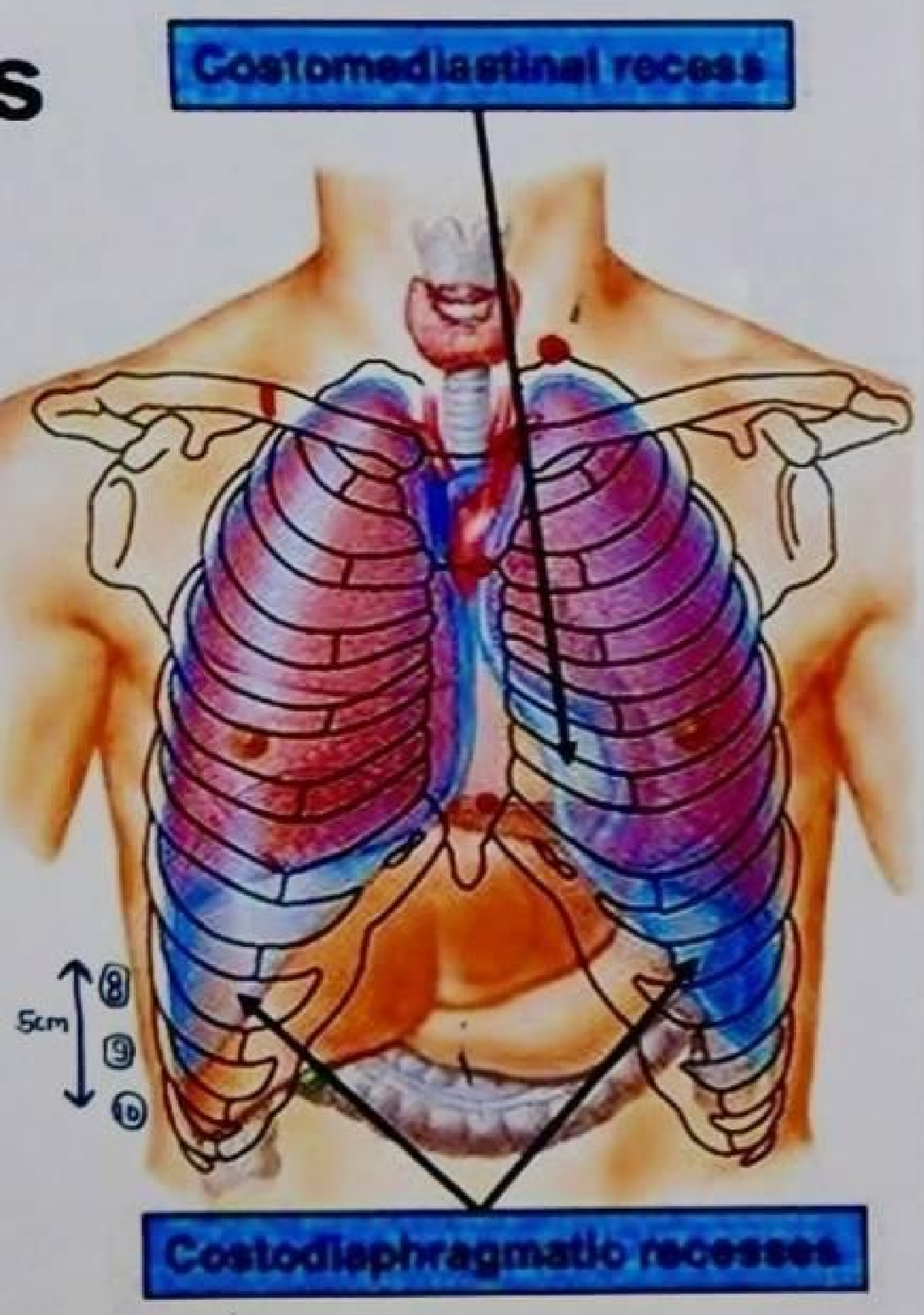
- Parietal pleura - surrounding root of lungs
  - Extend downward beyond root fold called pulmonary ligament

- fold contain - thin layer of loose areolar tissue
  - with few lymphatics

- Provide - dead space $\xrightarrow{\text{into which}}$ Pulmonary vein Expand ↓ during increased exercise venous return in

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Recesses of Pleura .....

↳ There are 2 recesses of parietal pleura

↳ act → reserve space for → lungs to expand during inspiration

Costomediastinal recess –

↳ lie anteriorly, behind sternum and costal cartilage

↳ b/w costal and mediastinal pleurae

↳ Particularly in relation → Cardiac notch of left lung of left lung

filled up by → anterior margin of lungs even during quiet breathing

only obvious → Region of cardiac notch of lung.

Costodiaphragmatic recess :-

↳ lie inferiorly, b/w costal and diaphragmatic Pleurae.

↳ Vertically → 5cm and extend from 8 to → 10 ribs in midaxillary line

filled up by → lungs expend into this recesses during inspiration

So these recess are obvious only in → expiration

↳ not in deep inspiration

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Fissures and Lobes of Lungs.....

↳ Right lung → divided into 3 lobes - upper, middle and lower by → Oblique fissure, horizontal fissure
↳ Left lung → divided into 2 lobes - upper and lower by → Oblique fissure
↳ Oblique fissure cut into whole thickness of lungs, except at hilum
↳ Due to oblique plane of fissure → lower lobe is more posterior
↳ upper & middle lobe more anterior

↳ Horizontal fissure - passes from anterior border of up to oblique fissure
- separate wedge shaped middle lobe from upper lobe
- Run horizontally at level of fourth costal cartilage

- Tongue-shaped projection of left lung below cardiac notch called Lingula

Impression on mediastinal surface :-

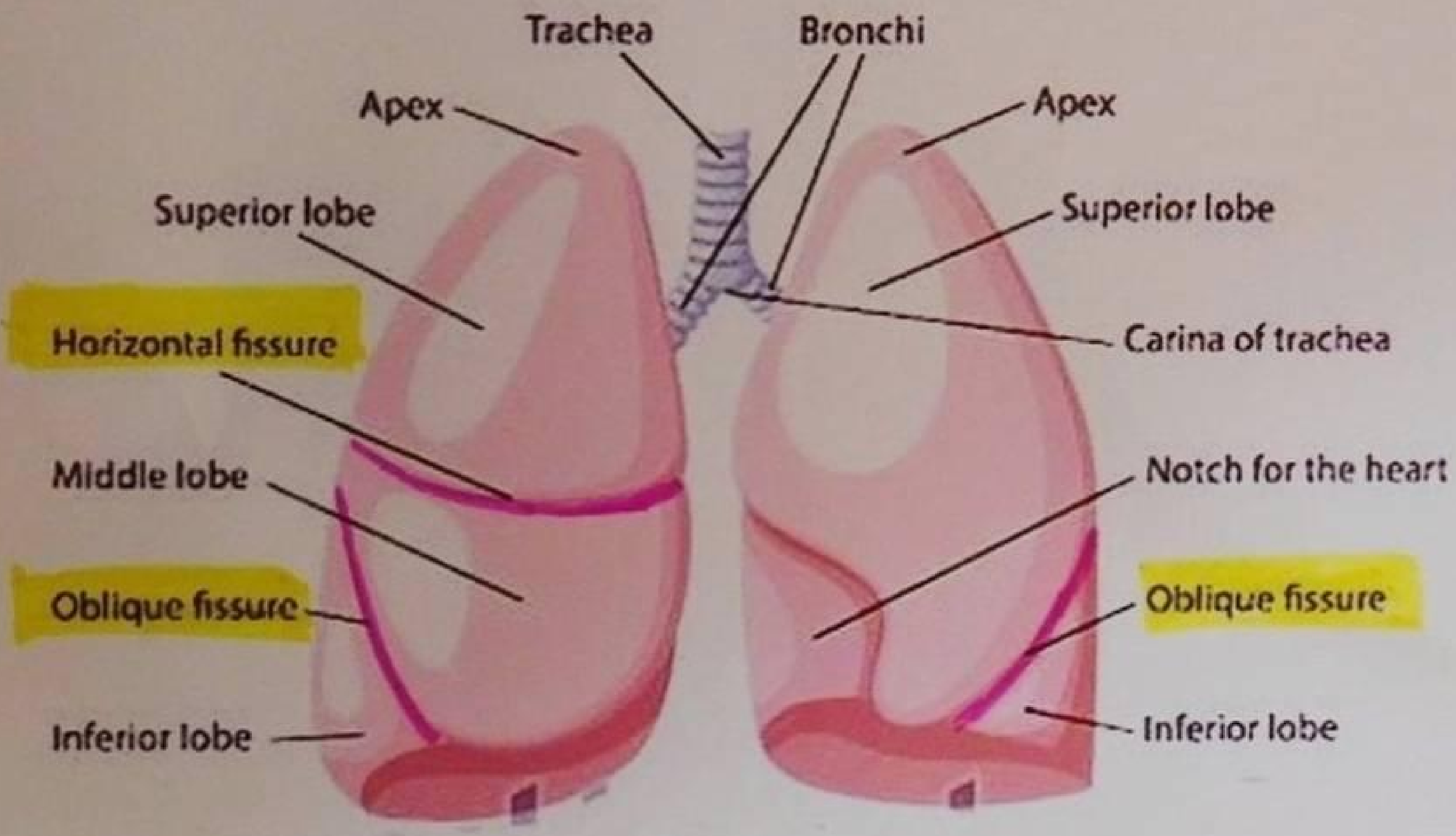
Right lungs - Oesophagus
- Trachea - Right vagus nerve
- Right phrenic nerve
- Superior vena cava → arch of azygos vein, Right brachiocephalic vein
- inferior vena cava
- Hilum
- Impression of right atrium

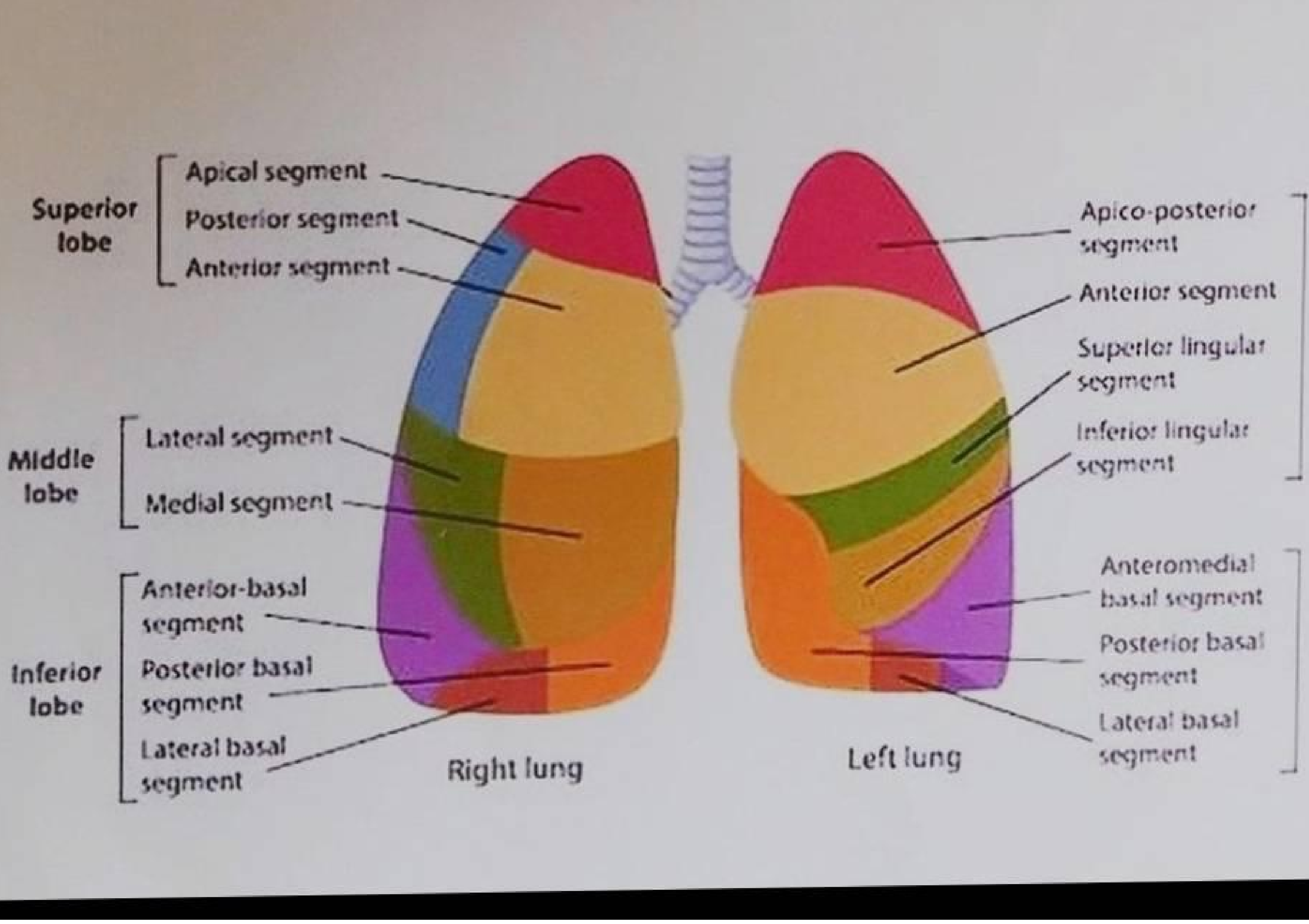
left lungs - Oesophagus - Thoracic duct
Left recurrent laryngeal nerve
Trachea
Left phrenic nerve
Left vagus nerve
Descending thoracic aorta
Pulmonary trunk
Hilum
Impression of left ventricle

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Lung lobes





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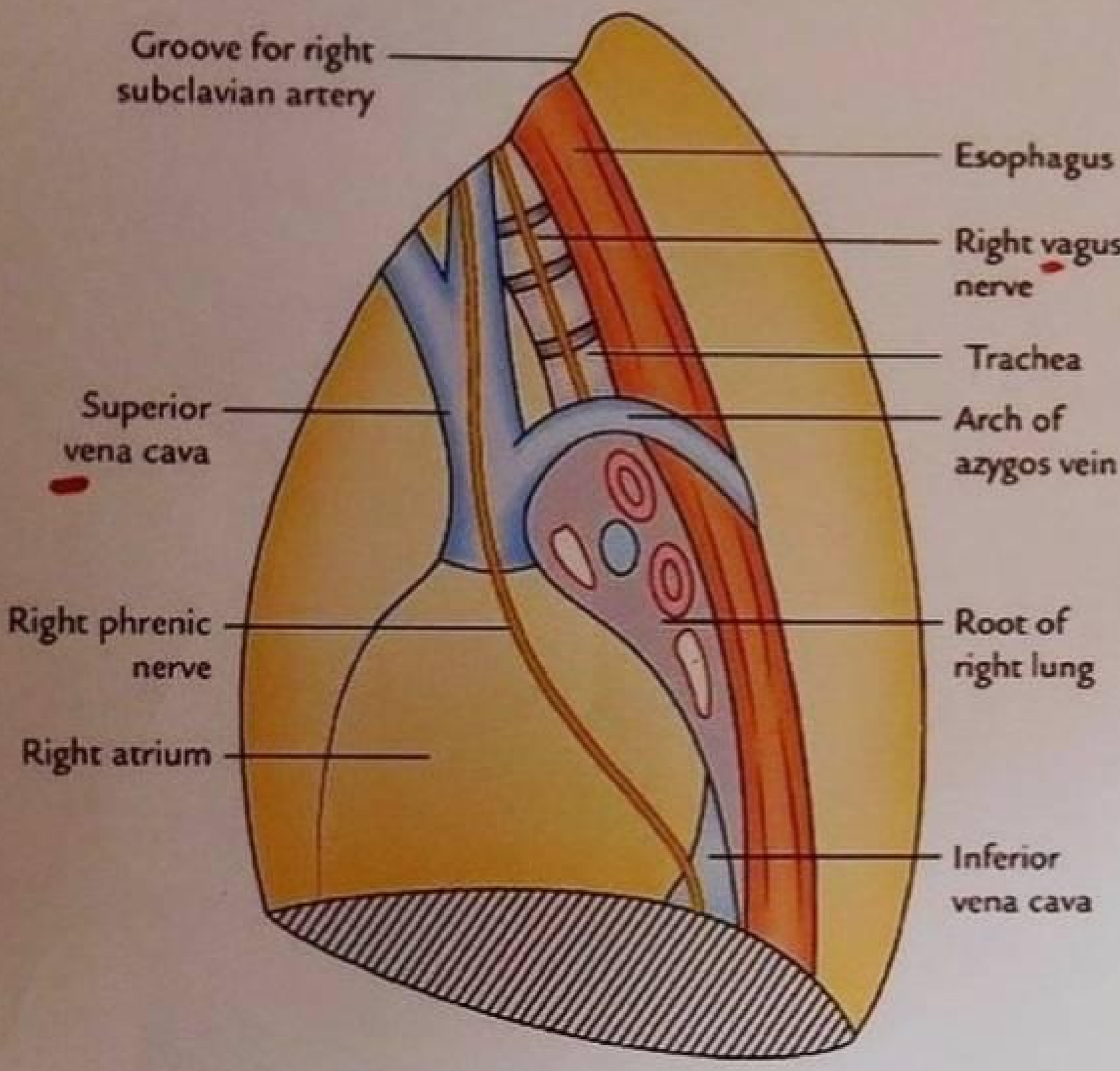
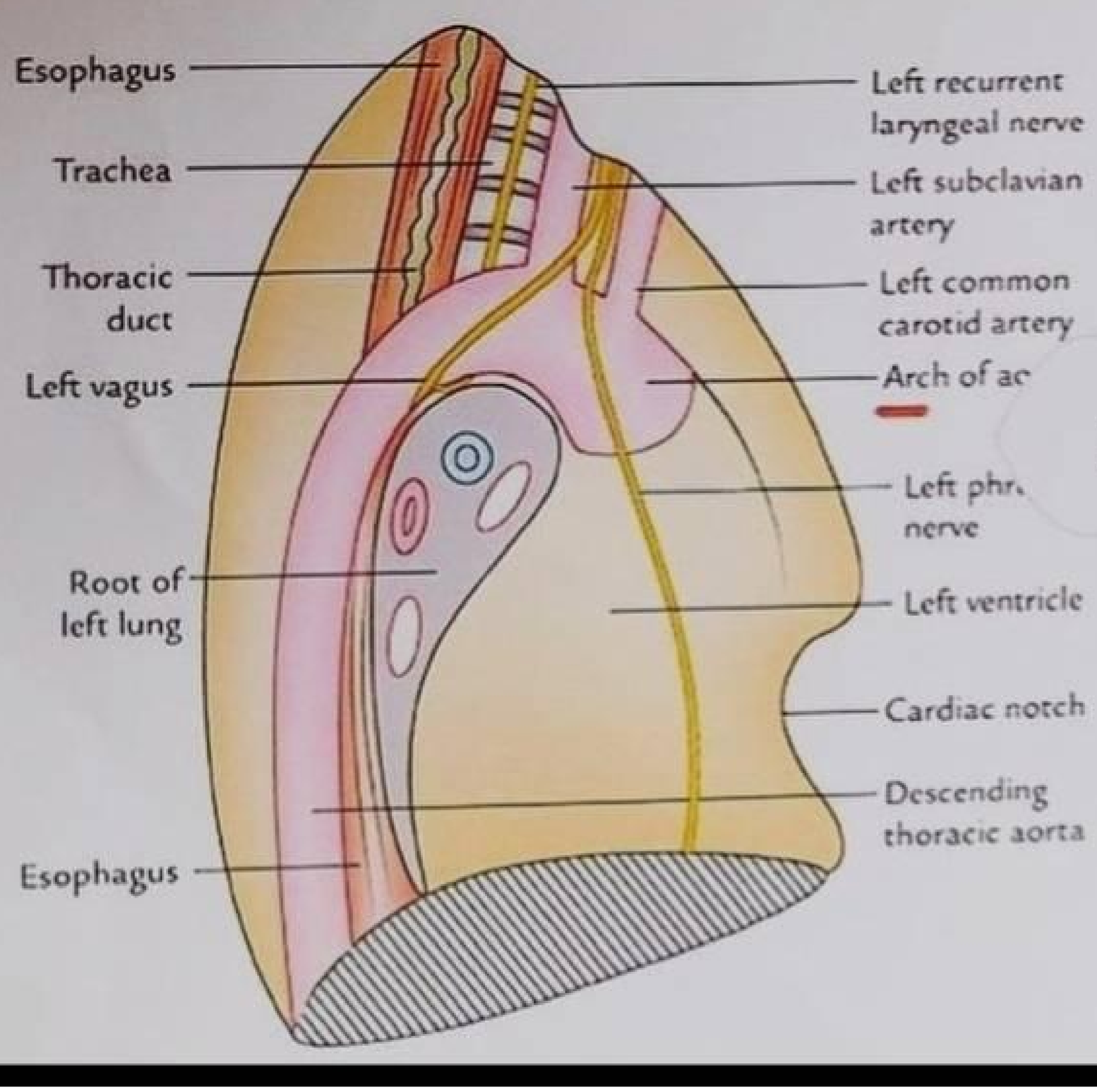
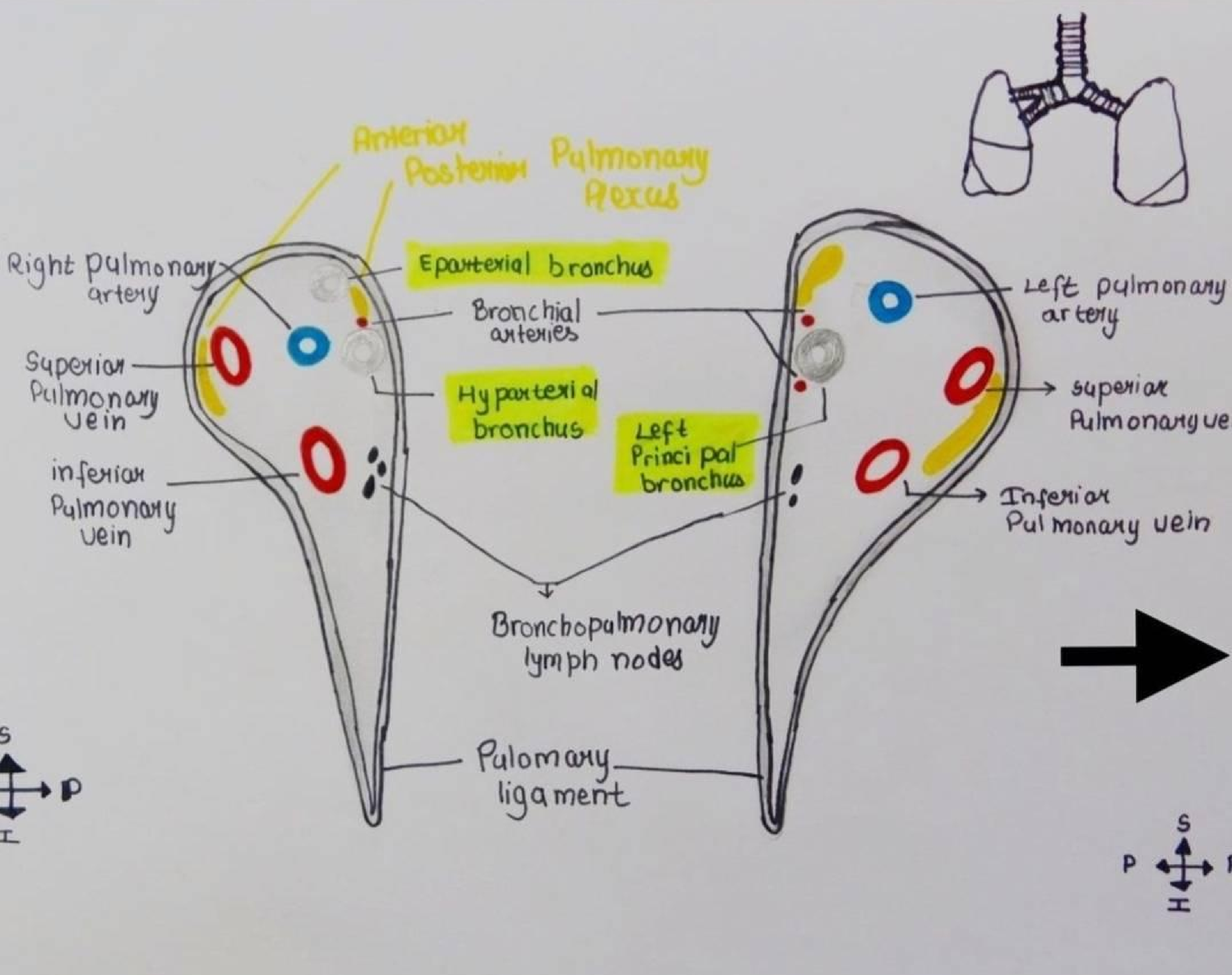


Fig. 18.3 Structures related to the mediastinal surface of the right lung.



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ROOTS OF LUNGS...

↳ Short, Broad pedicle which connect medial surface of lungs to mediastinum.

↳ Form by structure which either enter or come out of lungs at hilum.

↳ Lie opposite the body of 5th, 6th & 7th thoracic vertebrae.

Contents —

- Principal bronchus on the left side & eparterial & hyparterial bronchi on the Rt side.
- 1 pulmonary artery.
- 2 pulmonary veins, sup. & Inf.
- Bronchial arteries.
- Bronchial veins.
- Ant. & Post. pulmonary plexuses of nerves.
- Lymphatics of Lungs.

Arrangement of structures in the Root —

- Ant. to Post — similar in both sides.
  - * sup. pulmonary vein.
  - * Pulmonary artery.
  - * Bronchus.
- Above downward — Different in two sides.

Rt side.
Ep arterial bronchus.
Pulmonary artery.
Hyparterial bronchus.
Inf. pulmonary vein

Lt side.
Pulmonary artery.
Bronchus.
Inf. pulmonary vein.

Relations —

Ant. — Common on two sides
- Phrenic nerve.
- Pericardiacophrenic vessels.

on the Rt side.
Superior vena cava.

Post. — Common on two sides.
- Vagus nerve.

on Lt side.
- Descending thoracic aorta.

superior — on Rt side.
Terminal part of azygos vein

on Lt side.
Arch of aorta.

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Bronchopulmonary segments notes →



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→ 3 lobar bronchi - right side

6 → 2 lobar bronc left side

ar each bronchopulm y segment

↓ 10 or

right and side

ments....

left side

or of lungs

ward root of s

jacent p ary units

alo egmental vein to

ry accompany the

ccompany the bronchi ar

tal plane

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BRONCHIAL TREE...

↳ Trachea divides into 2 primary principal bronchi → Right principal bronchus
→ left principal bronchus

→ Each principal bronchus enter lungs through hilium

→ Divides into secondary lobar bronchi → one for each lobes → 3 lobar bronchi - right side
→ 2 lobar bronchi - left side

→ Each lobar bronchus divide into tertiary or segmental bronchi → one for each bronchopulmonary segment ↓ 10 on right and left side

→ Smaller branches are called respiratory bronchioles

Bronchopulmonary Segments....

→ There are 10 segment on right side and 10 on left side

Definition -
- well-defined anatomic, functional and surgical sector of lungs
- Each one is aerated by tertiary bronchus
- Each segment is pyramidal in space → Apex → toward root of lungs

→ Each segment has segmental bronchus
segmental artery
autonomic nerve

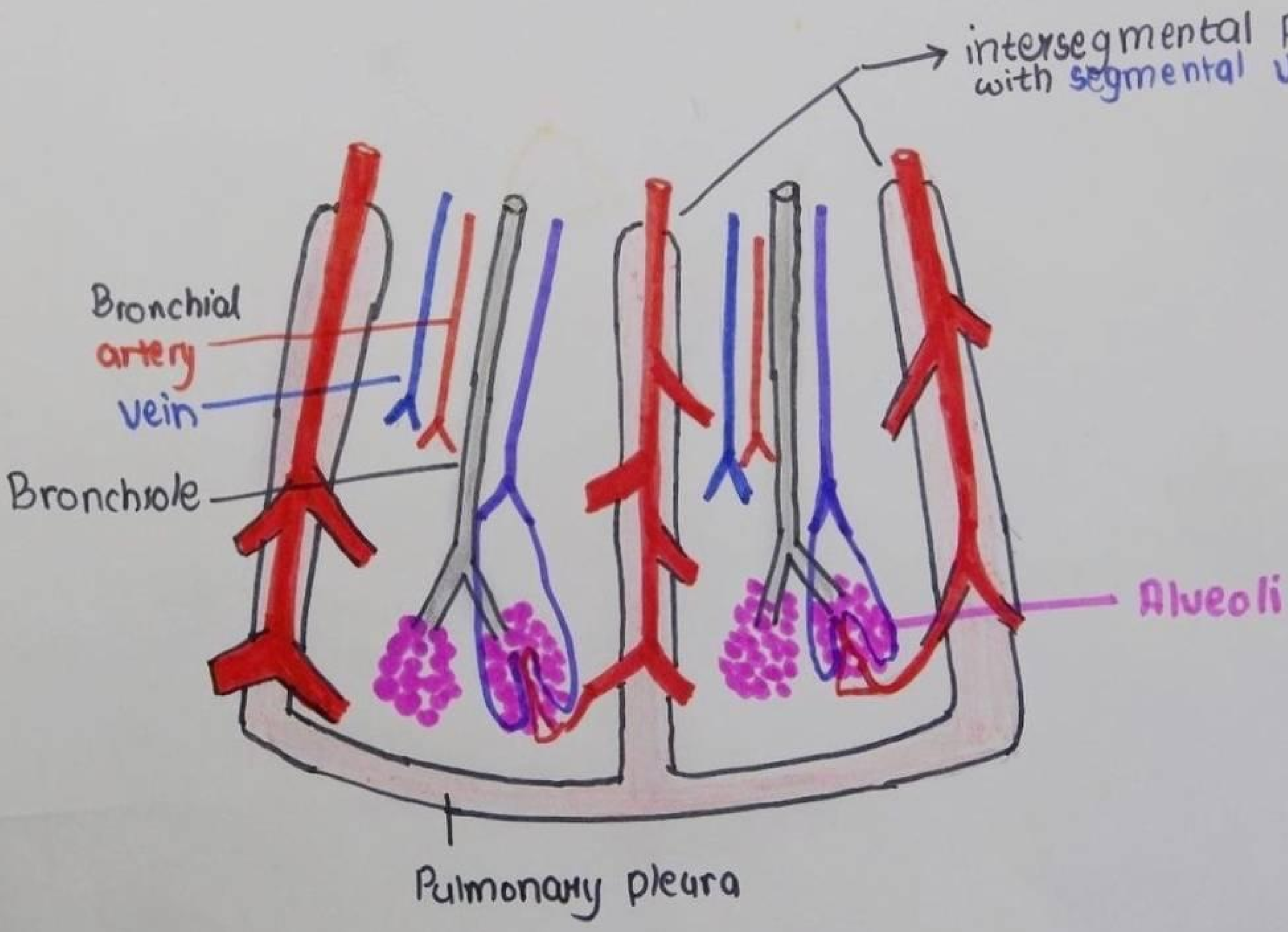
- Segmental venules lie in connective tissue b/w adjacent pulmonary units
- During segmental resection, the surgeon work along segmental vein to isolate a particular segment.

Relation to Pulmonary artery - Branches of pulmonary artery accompany the bronchi

Relation to pulmonary vein - Pulmonary vein do not accompany the bronchi or Pulmonary arteries
- Run in the intersegmental plane

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MEDIASTINUM...

↳ Middle space in the thoracic cavity blw lungs.
↳ Most imp content is heart enclose in pericardium.

Division :-

Sup. & Inf. mediastinum — Divided — Ant
Middle
Post.

Sup. mediastinum — separated from inf. by imaginary plane — passing through sternal angle anteriorly & thoracic vertebrae post.

Boundaries:-

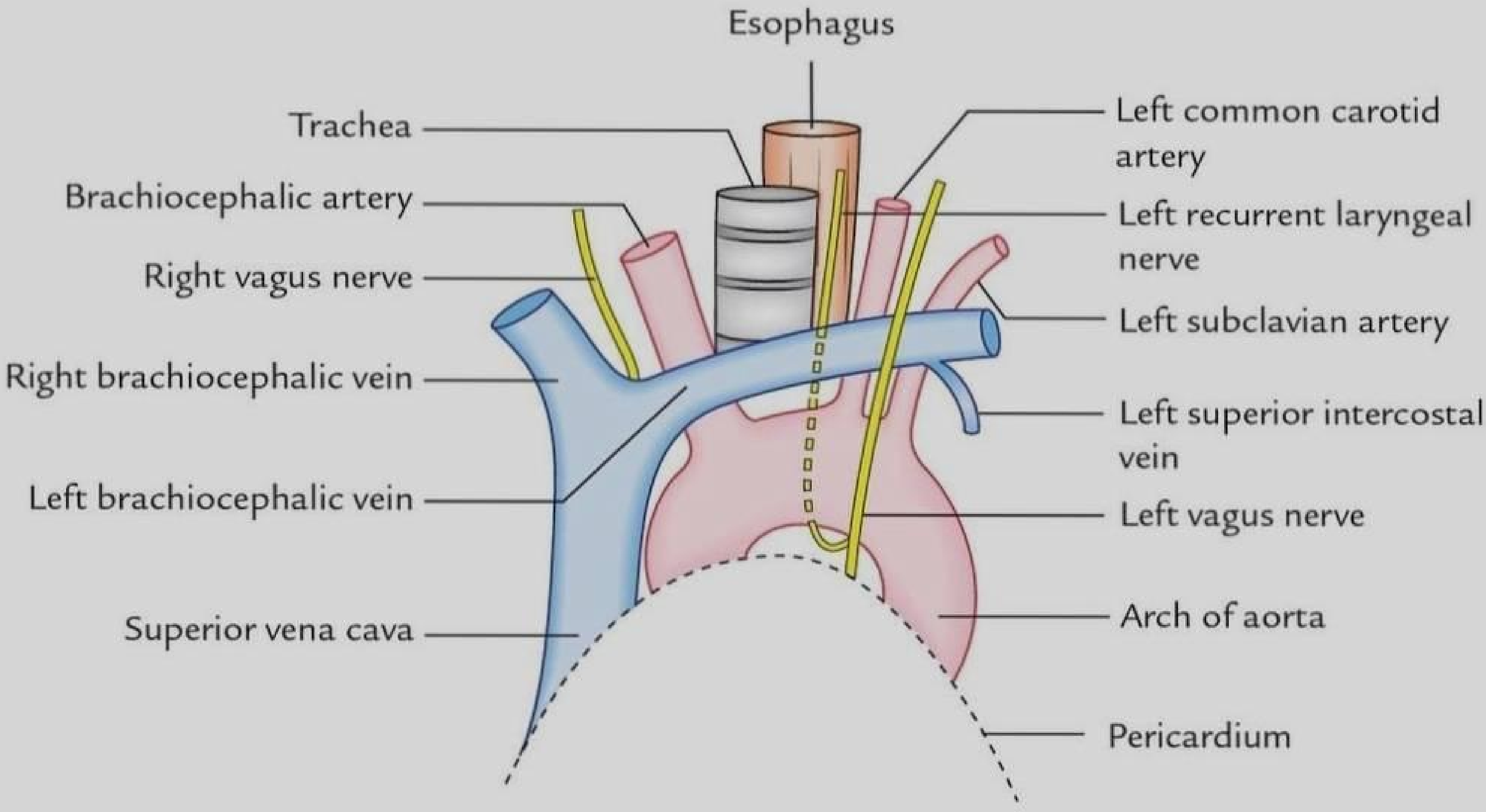
Ant. - Manubrium sterni.
Post. - Upper thoracic 4 vertebrae.
Sup. - Plane of thoracic inlet.
Inf. - An imaginary line.
on each side - Mediastinal pleura.

Contents :-

o- Trachea & Oesophagus.
o- Ms. - Sternohyoid.
Sternothyroid.
o- Arteries - Arch of aorta.
Brachiocephalic artery.
Lt. common carotid artery.
Lt. subclavian artery.
o- vein - Lt & Rt brachiocephalic vein.
Upper half of sup. venacava.
o- Nerves - Vagus.
Phrenic
Cardiac nerves of both sides.
o- Thoracic duct.
o- Lymph nodes.

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Inf. Mediastinum...

Divided into — Ant. — Middle — Post.

Ant. mediastinum — Very narrow space infront of pericardium.

Boundaries :-

Ant. — Body of sternum.
Post. — Pericardium.
Sup. — Imaginary line.
On each side — Mediastinal pleura.

Contents :-

o — sternopericardial ligament.
o — Lymph node.
o — Lowest part of Thymus.
o — Areolar tissue.

SLTA

Middle Mediastinum — occupied by pericardium & its contents.

Boundaries :-

Ant. — sternopericardial ligament.
Post. — Oesophagus.
Descending thoracic aorta.
On each side — Mediastinal pleura.

Contents :-

o — Heart enclosed in pericardium.
o — Ascending aorta.
o — Pulmonary trunk.
o — 2 pulmonary arteries.
o — Veins — Lower part of sup. venacava.
Terminal part of azygos vein.
Rt & Lt pulmonary vein.
o — Nerves — Phrenic
Deep cardiac plexus.
o — Lymph nodes.
o — Bifurcation of trachea.

Post. Mediastinum —

Boundaries :—

Ant. — Pericardium.
Bifurcation of trachea.

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Post. — Lower 8 thoracic vertebrae.
Each side. — Mediastinal pleura.

Content :-

- Oesophagus.
- Arteries — pescending thoracic aorta.
- vein — Azyos vein
  Hemiazyos vein
  Accessory hemiazyos vein.
- Nerve — Vagi.
  Splanchnic nerve.
  Greater, lesser & least arising from the
  8th lower thoracic ganglia of sympathetic chain.

- Lymph nodes.

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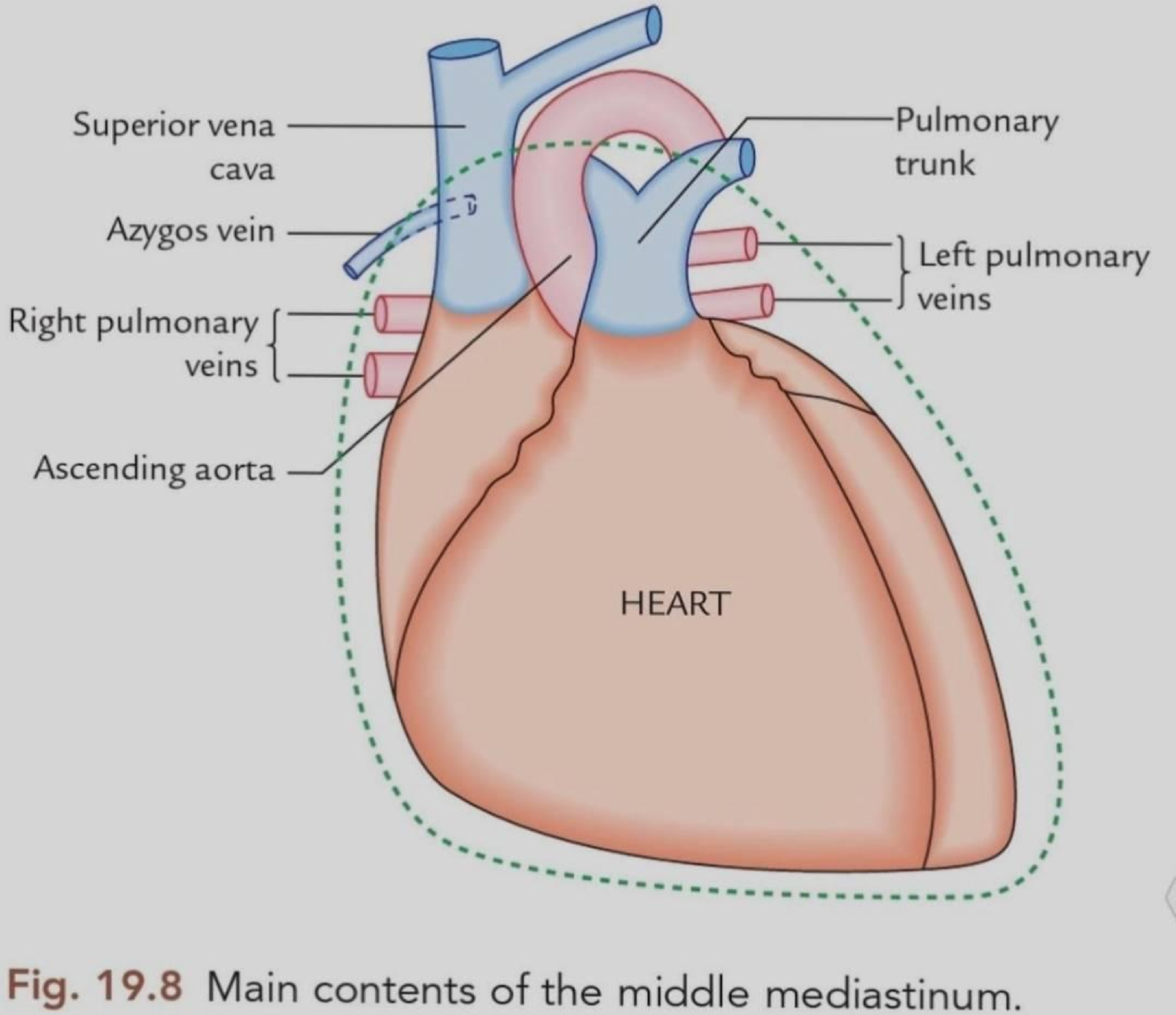
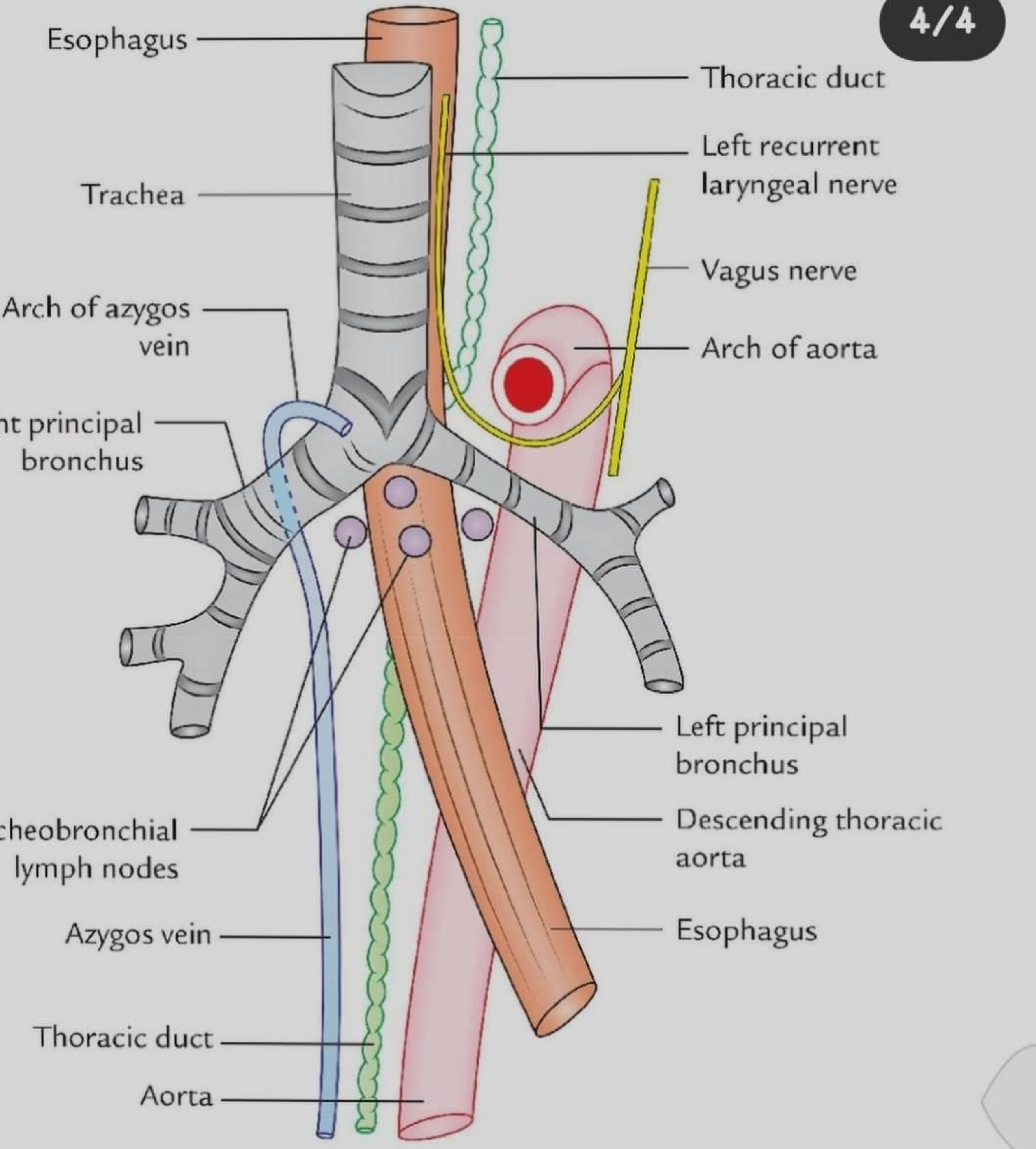
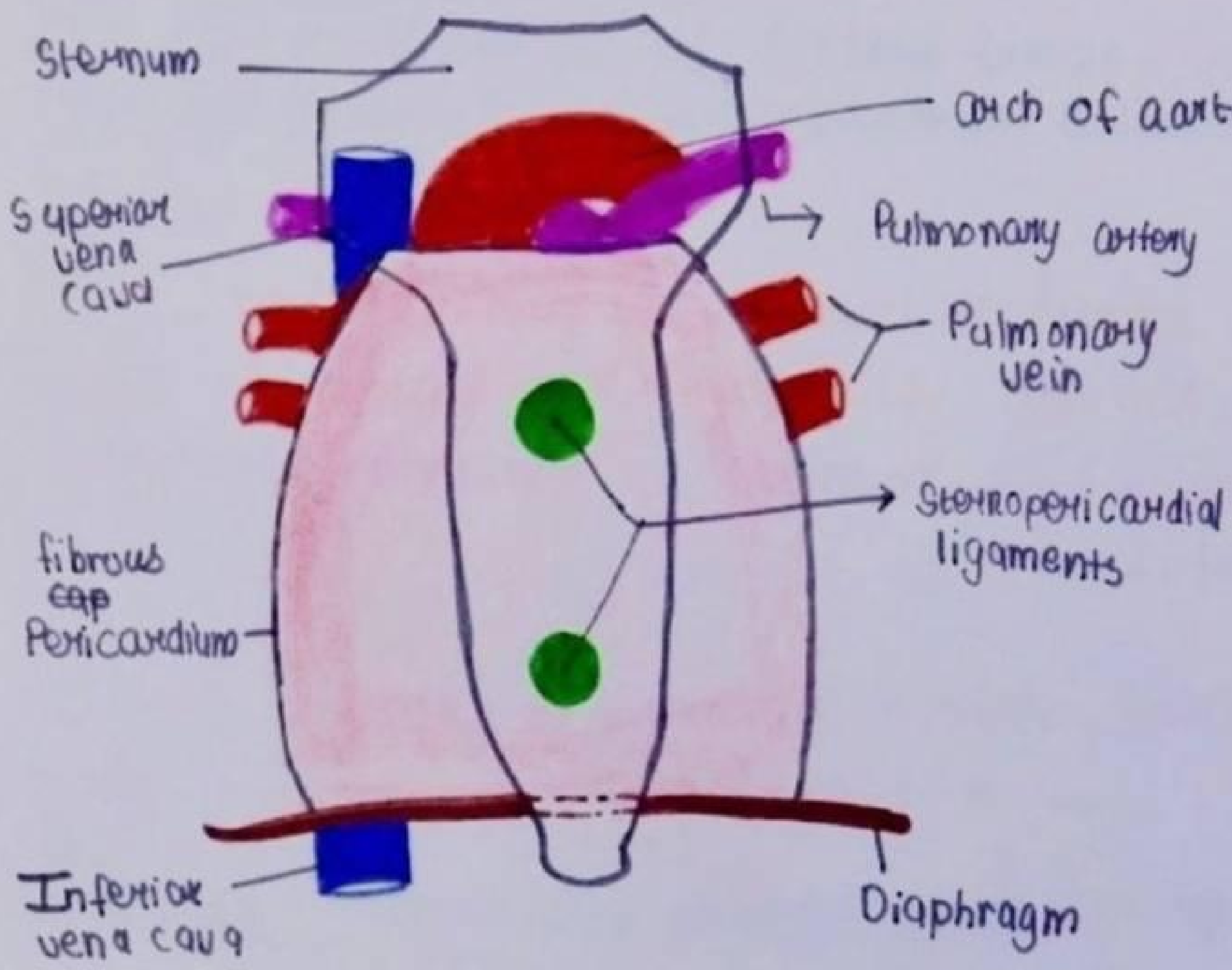


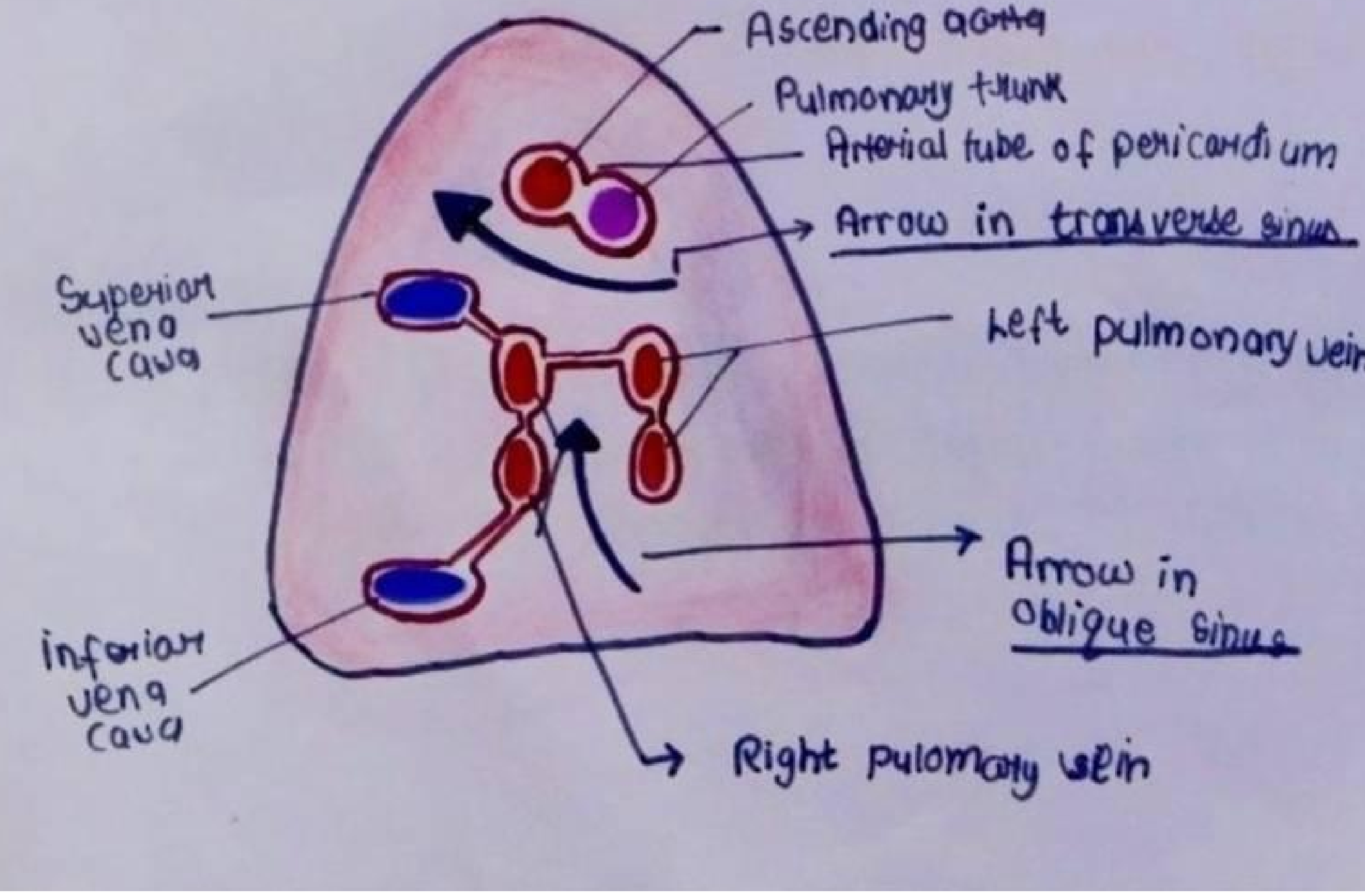
Fig. 19.8 Main contents of the middle mediastinum.

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PERICARDIUM...

↳ fibroserous sac which enclose the heart and root of great vessels.

→ Consist of fibrous pericardium and serous pericardium → Parietal pericardium
→ Visceral pericardium

Fibrous Pericardium -

↳ Conical sac made up of fibrous tissue

o- Parietal layer of serous pericardium attached to its deep surface

feature

- Apex is blunt and lie at level of sternal angle - fissued with great vessel
- Base is braad and inseparably blended with central tendon of diaphragm
- Anterior connected to sternum by → superior, Inferior → sternopericardial ligament
- Posterior related to principal bronchi, oesophagus with nerve plexus
- Each side by - mediastinal pleura

Blood supply - fibrous and parietal pericardia — supplied by → Internal thoracic, Musculophrenic arteries, Descending thoracic aorta

Vein drain into corresponding vein

Nerve supply - fibrous and parietal pericardia are supplied by - Phrenic nerve
- sensitive to pain

Serous pericardium -

↳ Serous pericardium is thin, double layered serous membrane

o- Outer layer - Parietal pericardium fused - with fibrous pericardium

o- inner - Visceral pericardium fused to heart

Pericardial cavity - potential space
- b/w parietal and visceral pericardium
- contains only thin film of serous fluid

Sinuses of pericardium -

↳ Epicardium at root great vessel arranged in form of two tube → arterial tube (ascending aorta, Pulmonary trunk ↑ enclose), venous tube (↓ enclose vena cava, Pulmonary vein)

↳ Passage b/w two tubes → Transverse sinus

↳ During development to begin → vein of heart crowded together
→ Heart increase in size and vein separate out
o- Pericardial reflection surround all of them and form Oblique pericardial sinus.

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↳ Epicardium at root great vessel arranged in form of two tube → arterial tube (↑ enclose)
                                                                → venous tube (↓ enclose vena cava, Pulmonary vein)

↳ Passage b/w two tubes → Transverse sinus

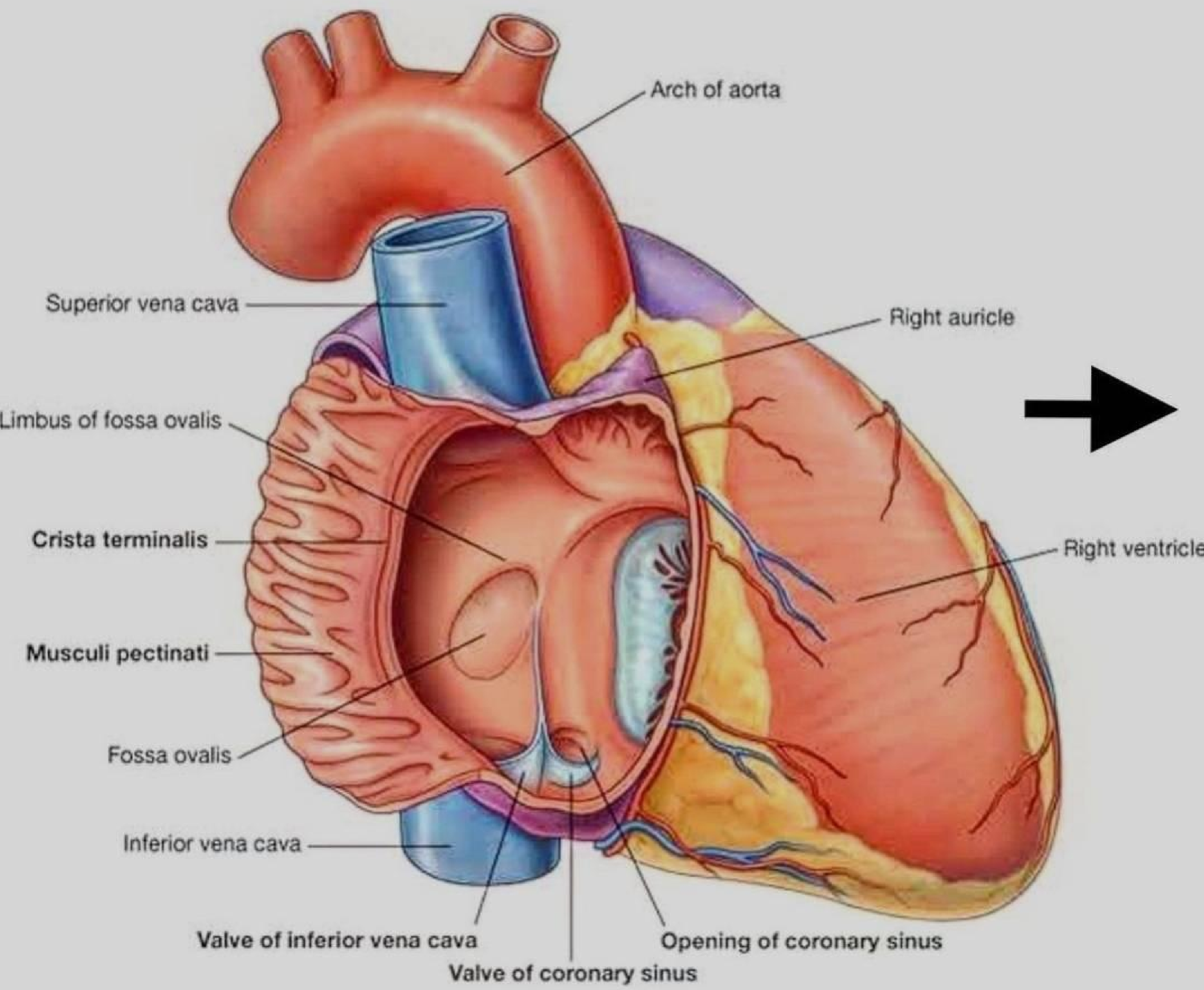
↳ During development to begin → vein of heart crowded together
→ Heart increase in size and vein separate out
o- Pericardial reflection surround all of them and form
Oblique pericardial sinus
↳ narrow gap behind the heart

Transverse sinus - bounded - anteriorly - ascending aorta
Pulmonary trunk
Posteriorly - Superior vena cava

Oblique sinus - Bounded - anteriorly - left atrium
Posteriorly - Parietal pericardium
oesophagus

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RIGHT ATRIUM ANATOMY NOTES



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Right atrium...

↳ Rt upper chamber of heart receive venous blood from body.
↳ Pump into Rt. ventricle through tricuspid valve.

External features :-

↳ Chamber is elongated vertically, receiving the sup. vena cava at the upper end & the inf. vena cava at the lower end.
↳ Along Rt border of the atrium, there is shallow ventrical groove which passes from the sup. vena cava to inf. vena cava. is cld sulcus terminalis.
↳ It produced by internal muscular ridge cld crista terminalis.
↳ Upper part of sulcus contained SA node.
↳ Atrioventricular groove separate Rt atrium from Rt ventricle.

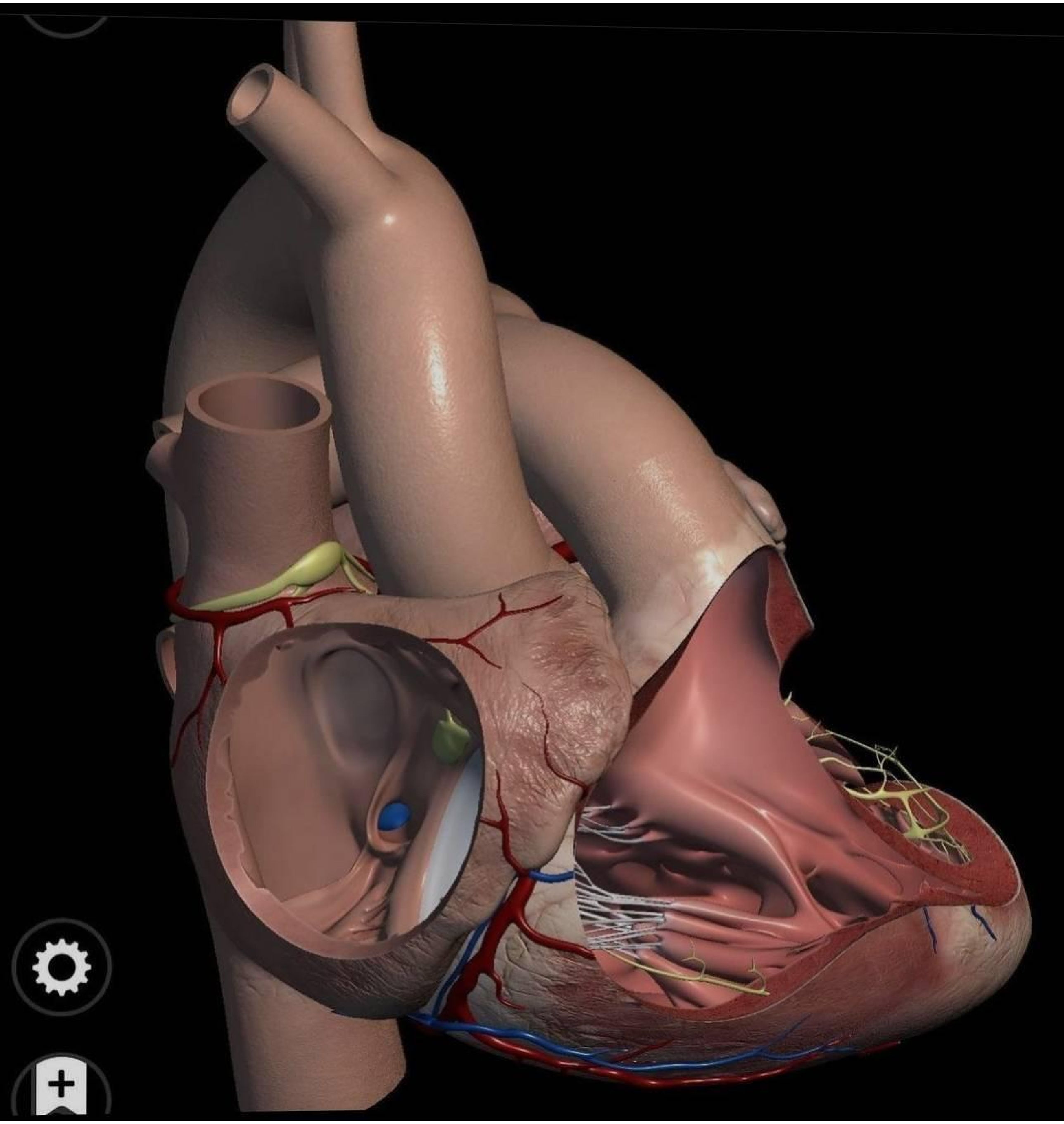
Inlet of Rt atrium :-

- sup. vena cava.
- Inf. vena cava
- coronary sinus.
- Ant. cardiac veins.

Internal Features :- Divided into 3 parts.

- Smooth part - Development - drived from Rt horn sinus venosus. Most of tributries except ant. cardiac vein open in it.
  * sup. vena cava.
  * Inf. vena cava — eustachian valve.
  * coronary sinus — Thebesian valve.
  * Venae cordis minimae - Numerous small veins present open in wall of all 4 chambers.
- Rough ant. part — Development - Drived from primitive atrioventricular chamber — It present a series of muscular Ridge. cld musculi pectinati.
- Internal septum — Development - Drived septum primum & septum secundum
  * It present the fossa ovalis a shallow scar depression in lower part.
  * Annulus ovalis & fossa ovalis the prominent margin of fossa ovalis

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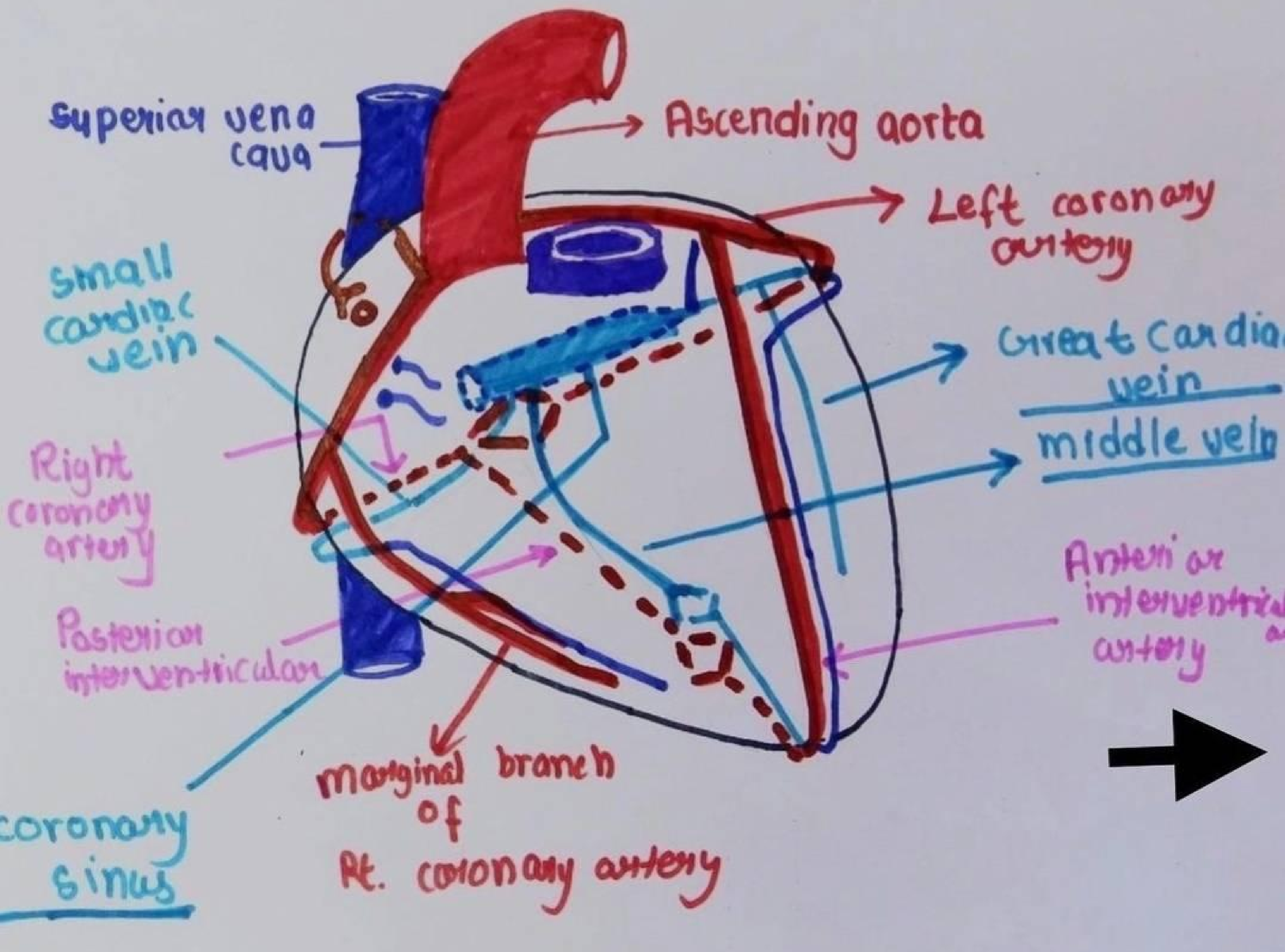


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Blood Supply of Heart



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Arterial Supply of Heart...

↳ Heart is supply by 2 coronary arteries – arising from ascending aorta

Rt Coronary arteries :- |

Position :- Smaller than Lt coronary arteries.
Arise from ant. aortic sinus.

Course :- First pass forward blw root of pulmonary trunk & Rt appicle auricle.
- Then run downward – Rt ant. coronary sulcus.
- winds around the inf. border to reach – diaphragmatic surface of heart.
- Terminate by anastomosis with Lt coronary artery – circumflex branch.

Branches :-
- Arterial branch – Ant, Post & Lat.
- Rt conus artery – Form circle around ~~coronary~~ pulmonary trunk. cld "Annulus of Vieussens."
- ventricular branch – Ant & Post group.
- Rt marginal artery.
- Post interventericular branch – lie in past interventricular groove.

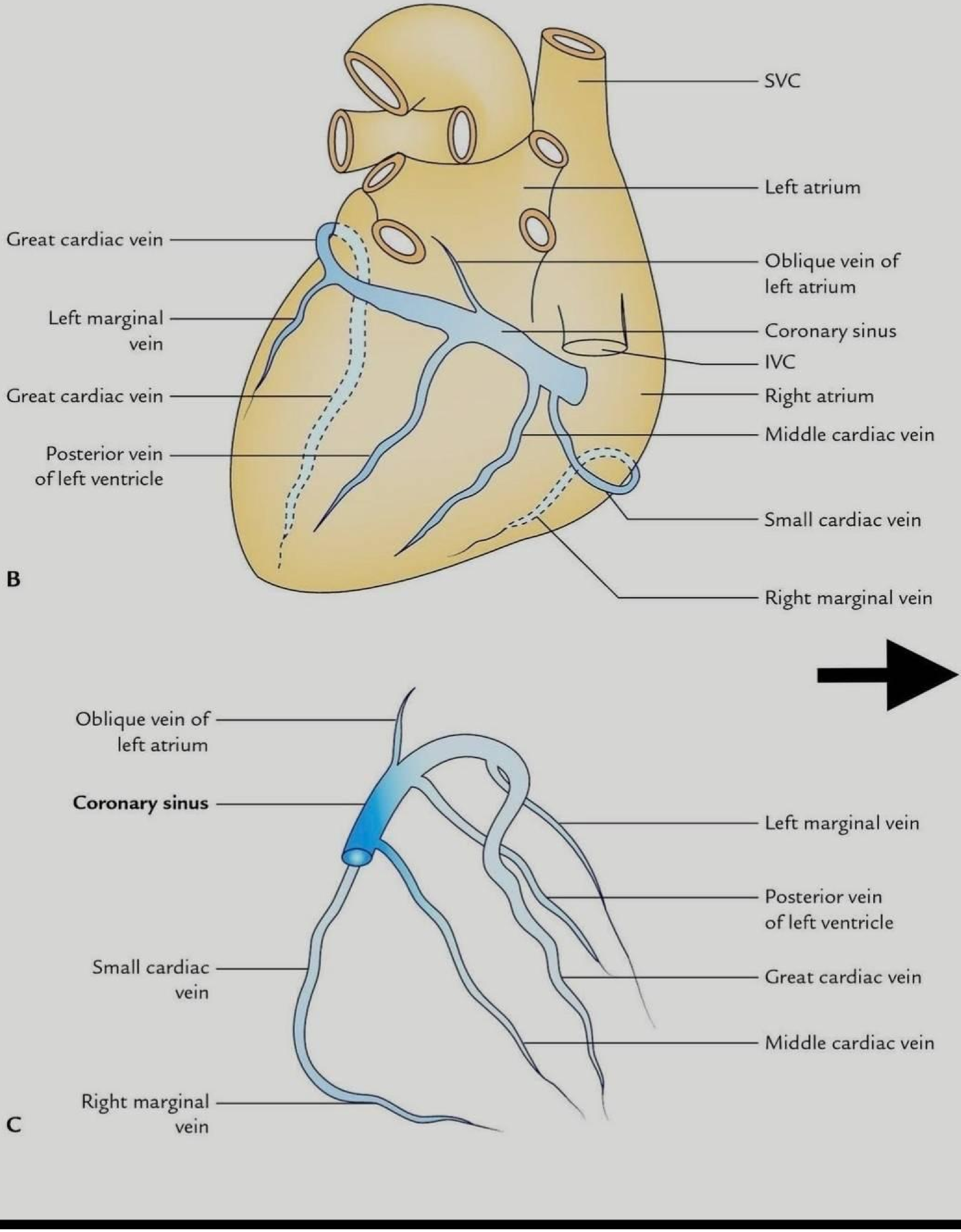
Area of Distribution :-
- Rt atrium
- ventricle – Great part of Rt ventricle – except area adjoing interventricular groove.
- Small part of Lt ventricle.
- whole conduction system of heart except a part of Lt branch of AV bundle.

Lt Coronary arteries :-

Position :- Arise from Lt aortic sinus.

Course :- First run forward to Lt end emerge blw pulmonary trunk & Lt auricle.
- Here it give the interventricular branch which run downward in groove of same name.
- Futher continuation LCA is cld circumflex artery.

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VEINS OF HEART...

↳ These are great cardiac vein, middle cardiac, Rt marginal vein, Post. vein of Lt ventricle, Anternal cardiac vein, Venae cordis minimae

↳ All vein except = ant. cardiac open into coronary sinus — Rt atrium.
venae cordis minimae.
↳ open - directly into Rt atrium.

Coronary Sinus :- Largest vein of heart.
- Situated in Lt post. coronary sulcus. end into Rt atrium.
- Tributries — 1) Great cardiac vein — accompany first Ant. ventricular artery than Lt coronary artery.

2) Middle cardiac vein — Accompany — Post. interventricular artery.

3) Small cardiac vein — Accompany - Rt. coronary artery.

4) Rt Marginal vein — Accompany marginal branch of Rt. coronary artery.

5) Post vein of Lt ventricle — Run on diaphragmatic surface on Lt ventricle.

6) Oblique vein of Lt atrium — Running of post. surface Lt atrium.

Ant. cardiac. vein :— 3 4 Small vein which run (Parallel) // to each other. On the ant. wall Rt ventricle
- Open directly into Rt atrium.

Vena cordis minimae :- Numerous small valvular vein present in all four chamber of heart - Directly open Rt atrium.

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Upper limb - 2

INTERMUSCULAR SPACE...

↳ It is formed by 3 Ms.

o— Long head of tricep., Teres minor & major.

o— Long head of tricep $\xrightarrow{\text{origin}}$ scapula = Infra glenoid tubercule.
$\xrightarrow{\text{insertion}}$ Ulna = olecron

o— Teres minor $\xrightarrow{\text{origin}}$ scapula = lateral border.
$\xrightarrow{\text{insertion}}$ Humerus = greater tubercule.

o— Teres major $\xrightarrow{\text{origin}}$ scapula = Infior angle
$\xrightarrow{\text{insertion}}$ Biccpital groove.
medial lip

↳ These Ms. form 3 spaces — Quatrianglar space
* Lower triangular space.
* Upper triangular space.

① **Quatriangular space** :—

Boundaries — superior ↱ Inf. border of teres minor.
Subcapular in front.
Inferior (superior) border of teres major.
Medial — Lat. border of Long head of triceps brachi
Lat. — surgical neck of humerus.

Content — Axillary nerve.
Post. circumflex humeral vessels.

② **upper triangular space** :—

Boundaries — sup. — Inf. border of teres minor.
Lat. — Med. border of long head of triceps brachi.
Inf — sup. border of teres major.

Content — Circumflex scapular artery.
Suprascapular artery.

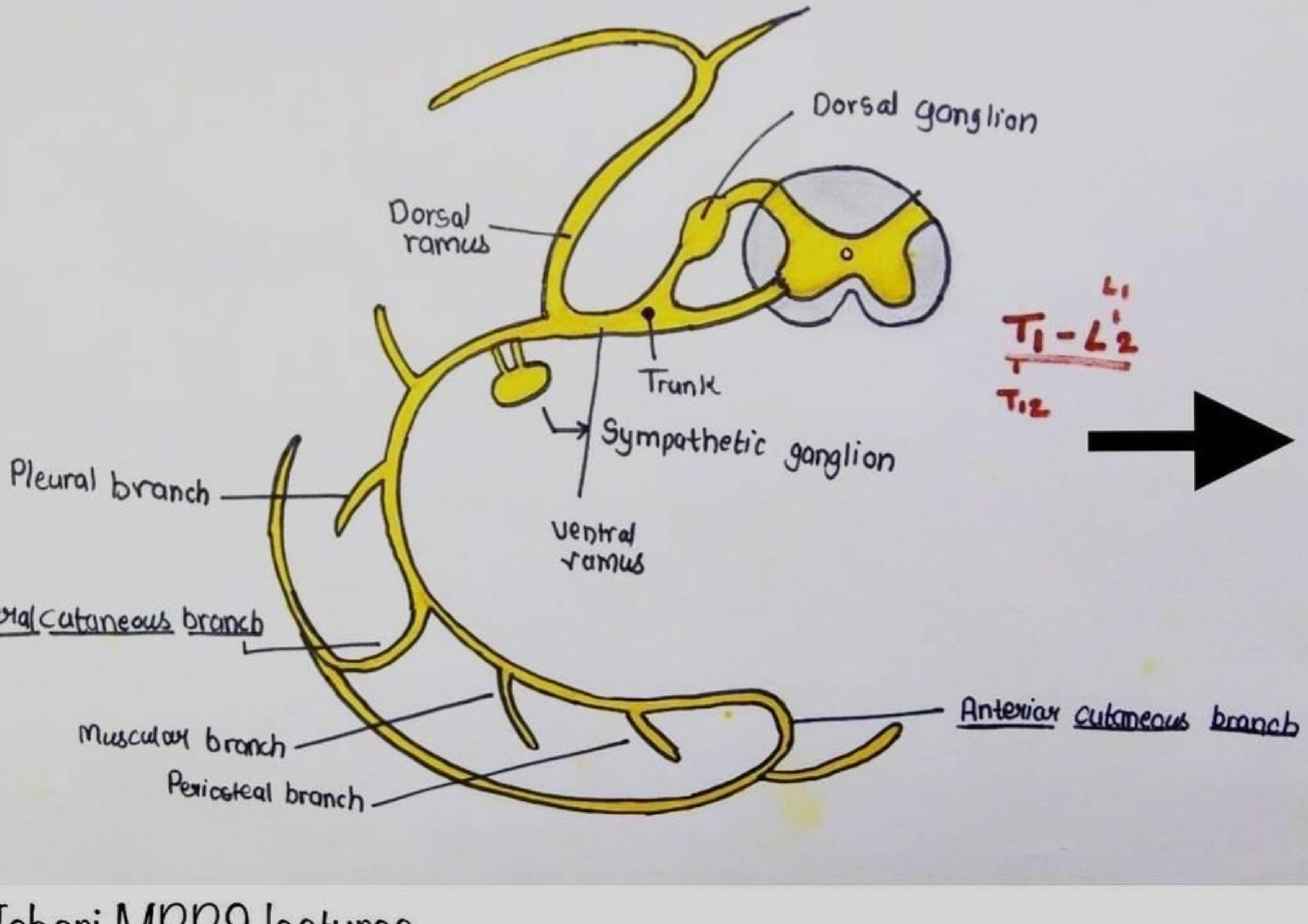
③ **Lower [redacted] triangular space** :—

Boundaries — Med. — Lat. border of long head of triceps brachi
Lat. — Med. border of humerus.
Sup. — Lower border of teres major.

Content — Radial nerve.
Profunda brachii = vessels.

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Intercostal Nerve Notes



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INTERCOSTAL NERVE..

↳ All intercoastal nerve — supply intercoastal Ms.

Intercoastal Nerve :-

o- Intercoastal Nerve are ant. primary rami of thoracic 1-12 Spinal nerve.

o- The ant. pri rami of 12 thoracic nerve — Form subcoastal nerve.

In additional to supply intercoastal space -

$T_1 \cdot T_2$ o- upper 2 IN - also supply upper limb.

o- Lower 5 - supply abdominal wall.
Also cld thoracoabdominal nerve.

o- 3-6 supply only thoracic wall they are cld typical intercoastal N.

Course :-

o- IN run in coastal groove.

o- And near sternum.

Branches :-

o- Muscular branch — Numerous muscular branch supply intercos
-tal ms — Transversus thoracis.
Serratus past. sup.

o- sensory branch - Main branch & colateral branch. also supply paretial pleura & periosteum of rib.

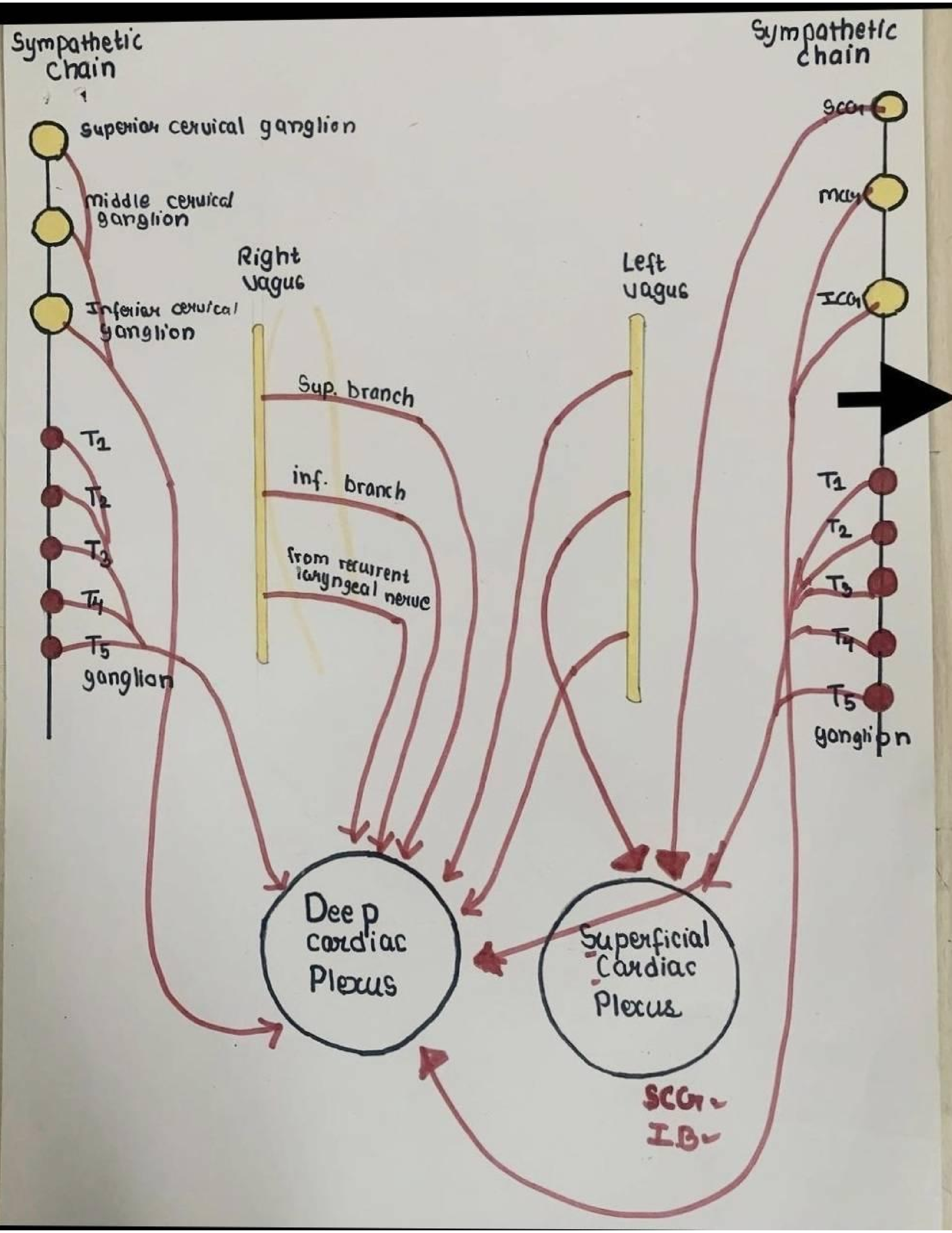
o- Lat. cutaneous branch arise — near angle of rib.
where it pair Intercoastal Ms. & distributed to the skin.

o- Ant. cutaneous branch — Emerge on the side of sternum.
— Supply overlying skin.

o- communicating branch — Each nerve is connected to thoracic sympathetic ganglion by a distal place white & proximal grey matter communicans.

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Nerve supply of Heart...

- Parasympathetic nerve **reach** → via **Vagus nerve**
  ↳ Cardioinhibitory - slow down heart rate

- Sympathetic nerve → derived from T_1 - T_5 ganglion
  ↳ Cardio-accelerotory - inc heart rate

- Both Sym. and paraSym. form — Superficial Cardiac plexuses
  — Deep Cardiac plexuses

- Superficial cardiac plexus - Below arch of aorta in front of R. pulmonary artery.

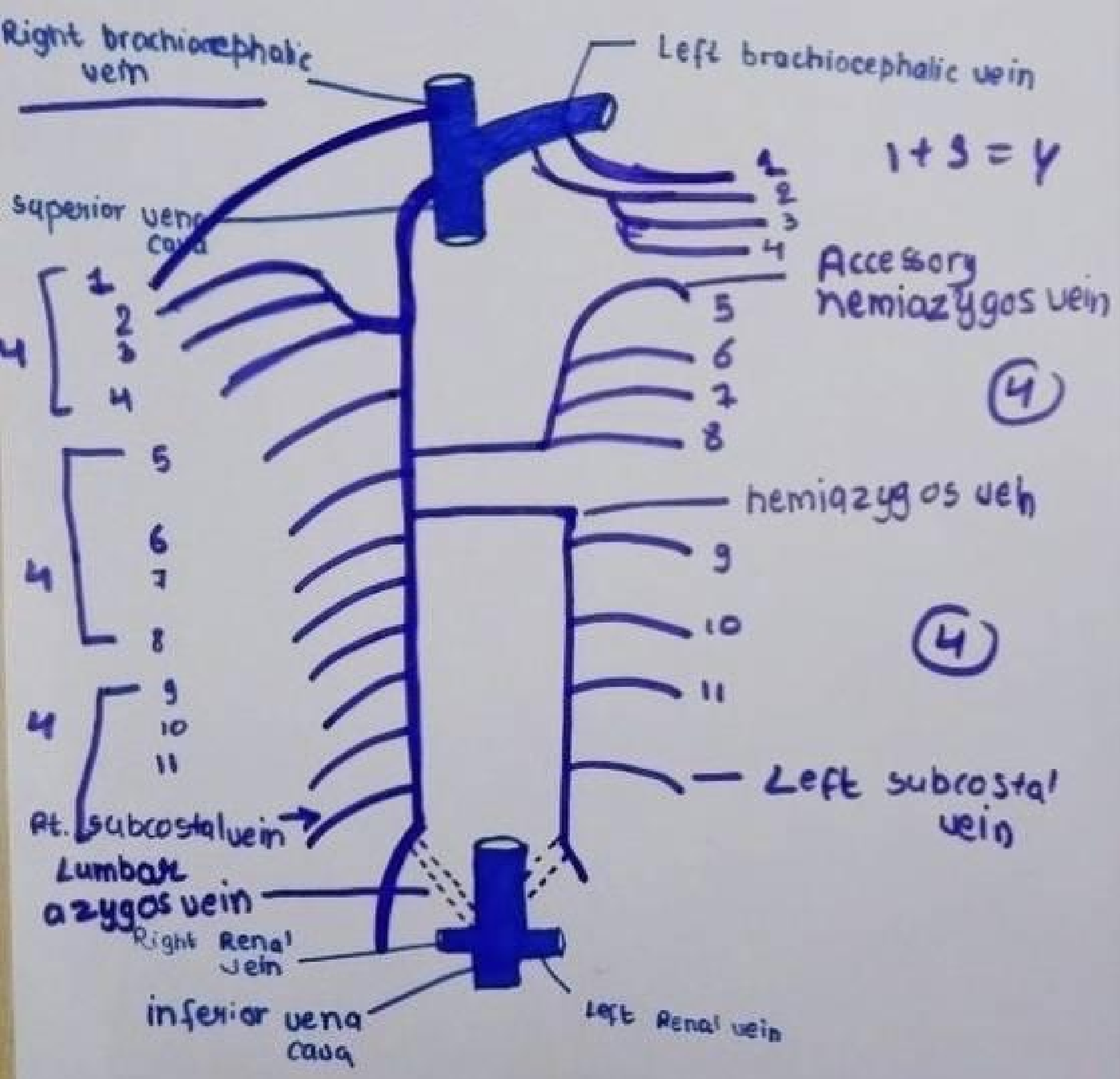
  - form by - Superficial cardiac plexus → Superior cervical ganglion - Left Sym. Chain

  → Inferior branch - Left Vagus

- Deep cardiac plexus - front of bifurcation of trachea
  ↳ derived from - all cervical and T_1 - T_5 ganglion
  - cardiac brach of Vagus nerve

↳ except those which form superficial plexus
↓
o — SCG - Left Sym. chain
o - inferior branch - Left Vagus

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Azygos vein...

↳ The azygos term means unpair.
↳ Azygos vein drain thoracic wall & the upper lumbar region.
↳ From lmp. channel connecting IVC
↳ vein occupies upper part of post wall of abdominal wall & the post. mediastinum.

Formation :-

↳ Form by union of lumbar azygos
- Rt sub-costal.
- Rt ascending lumber vein

Tributaries :-

↳ Rt sup. intercoastal vein —Form→ by union of 2nd 3rd 4th post. intercoastal vein.
- 5-8 post intercoastal vein.

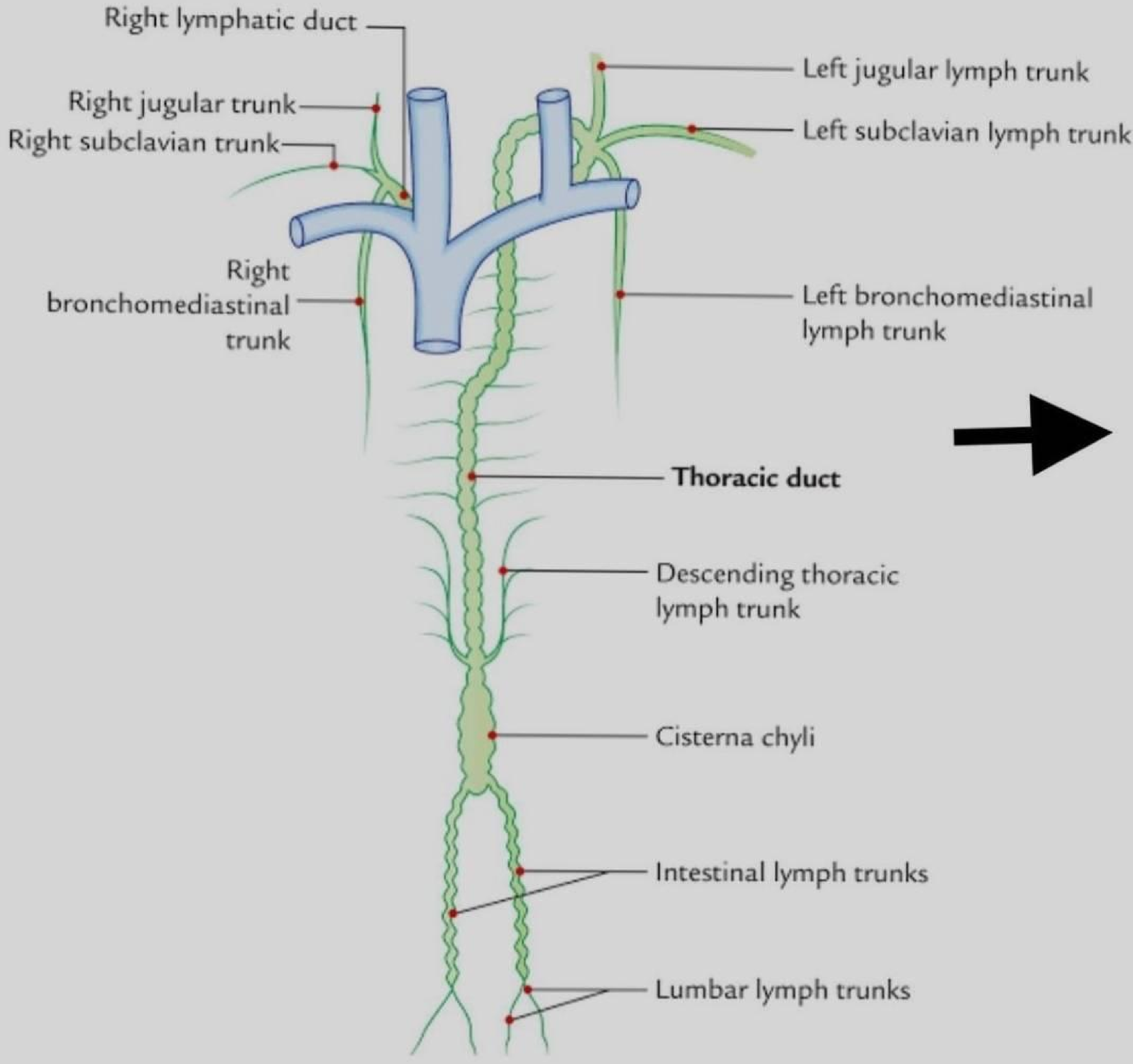
↳ Hemiazygos ⟶ at level of lower border of 8 thoracic vertebrae.

Tributaries :-
9-11 post intercoastal vein.

Accessory hemiazygos vein — At the level of upper border of 8th thoracic vertebrae.

Tributaries :-
5-8 ~~post~~ left post. intercoastal vein.
several oesophageal, mediastinal, pericardical vein.
Rt & Lt sub coastal vein.

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THORACIC DUCT...

↳ Largest lymphatic vessel of the body.
↳ Extend from upper part of abdomen to lower part of neck.
↳ Cross post. border, sup part of mediastinum.

Course :—

- Begins as a continuation of the upper end of the cisterna chyli. near lower border — 12 thoracic vertebrae.
- Enter thorax through aortic opening of diaphragm.
- Then ~~descend~~ ascend through post. mediastinum from level of 12th — 5th thoracic vertebrae.
- Then it course through sup. mediastinum.
- In neck — arch lat. at level of 7th cervical vertebrae.
- And by opening into angle of junction blw lt subclavian & lt internal jugular vein.

Relation :—

- At aortic opening of diaphragm :—

  Ant. — Diaphragm.

  Post. — vertebral colum.

  To the right - Azygos vein

  To the left — Aorta

- In the post. Mediastinum :—

  Ant —
  - Diaphragm
  - Oesophagus.
  - Rt pleural recess.

  Post. — Vertebral column
  - Rt post. intercostal arteries.

  the right — Azygos vein

  To the left - Descending aorta

- In the sup. mediastinum.

  Ant - Arch of aorta

  Post - Vetebral column.

  To the right - Oesophagus

  To the left - Pleura.

- In the neck :—

  The thoracic duct forms an arch rising about 3-4 cm above the clavicle.

  Ant — Lt common carotid artery.
  Lt vagus.
  internal jugular vein

  Post. — vertebral artery.
  sympathetic trunk.

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Tributaries :—

Thoracic duct receive lymph from both half of body below diaphragm. & Lt half above abdomen.

o- In thorax thoracic duct receive — lymph vessel from post. mediastinal node. & small intercoastal node.

o- Lt jugular trunk, Lt subclavian trunk end in thoracic duct

o- Lt bronchiomediastinal trunk — drain lymph from Lt half of thorax & end in thoracic duct.

Rt half above diaphragm :—

Rt side their is lymphatic duct

o- Into which Rt bronchiomediastinal
Rt jugular.
Rt subclavian lymph drain.

o- Rt lymphatic trunk end in Rt brachiocephalic vein.

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Covering of the Thoracic Wall :-

F-1

Thoracic wall is covered from outside to inside by the following structure - Skin, superficial fascia, deep fascia and extrinsic muscle.

Extrinsic muscles :- Muscle of Upper limb

- Pectoralis major
- Trapezius
- Serratus anterior
- Pectoralis minor
- Latissimus dorsi
- Rhomboid major
- Rhomboid minor
- Serratus posterior superior
- Serratus posterior inferior

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Muscles of Abdomen

- Rectus abdominis
- External oblique

Intercostal Spaces :-

↳ Gaps b/w ribs are called intercostal spaces filled by intercostal muscle and contain - intercostal nerve, vessel

9 - intercostal space - anteriorly
11 - intercostal space - Posteriorly

Intercostal muscle :-

↳ External intercostal muscle
- Internal intercostal muscle
- Transversus thoracis muscle → divied 3 parts → Subcostalis, Intercostalis intimi, Sterno costalis

Extend

↳ External intercostal muscle - Posteriorly - tubercle of ribs
Anteriorly - costochondral junction

o → Anteriorly - B/w costochondral junction and sternum → replaced by → External intercostal membrane

o - Posteriorly - end of muscle is continuous with posterior fibres → superior costotransverse ligament

Direction of fibres - Downward, forward and medially in front

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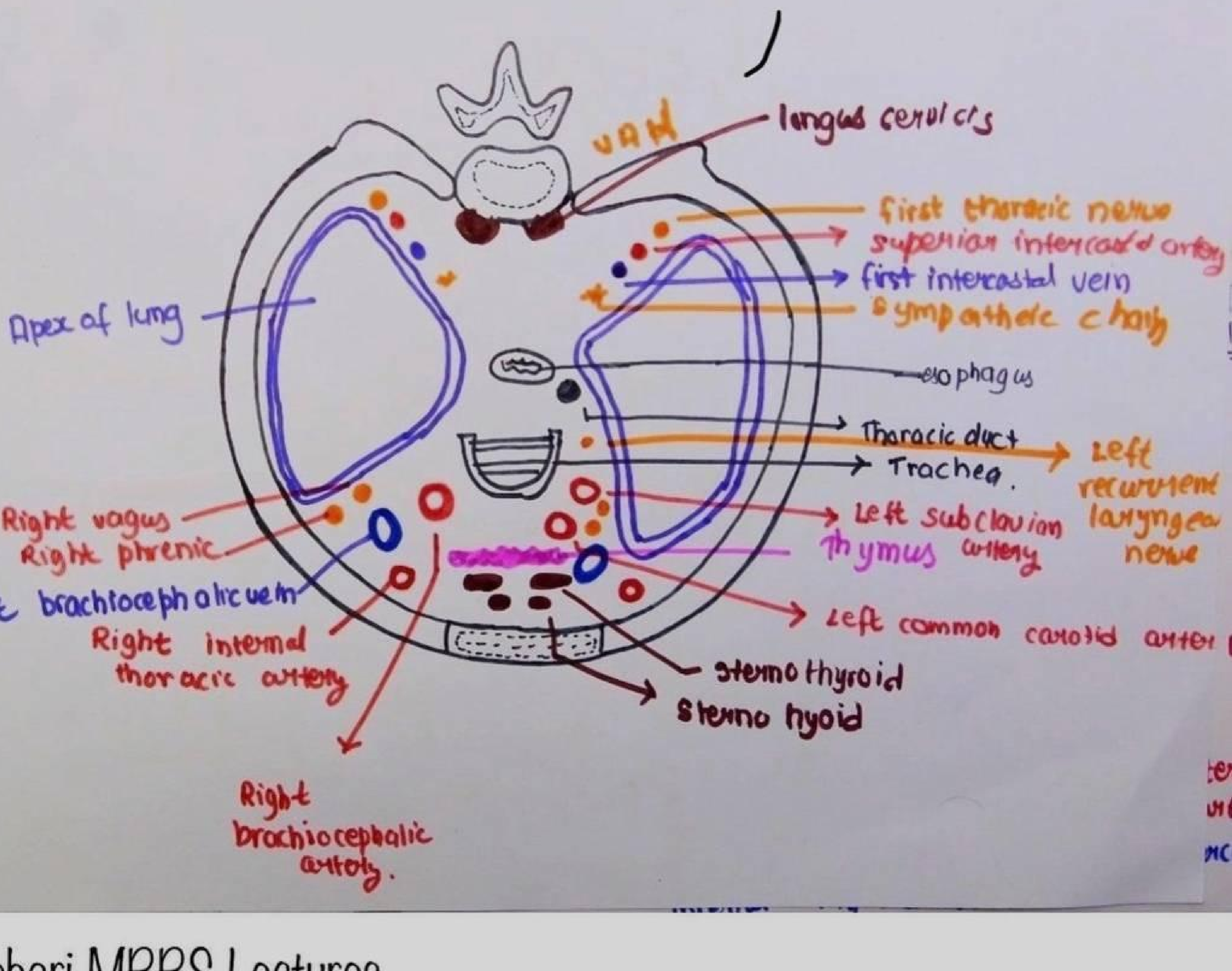
internal intercostal muscle - Anteriorly - lateral border of sternum → continuous superior costo-transverse liga.
Posteriorly - angle of rib → Beyond Posterior intercostal membrane

↳ Direction of fibres - downward backward, laterally

o- Subcostalis - Posterior part of the lower intercostal space.
o- intercostalis - middle 2/4 of all intercostal space
o- sternocostalis - Anterior part of upper intercostal space
↳ same direction as internal intercostal muscle

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Inlet Of Thorax Notes



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INLET OF THORAX

T-2

- Narrow upper end of thorax - continuous with the neck
- Kidney - shaped

Boundaries -

Anteriorly :- Upper border of manubrium sterni
Posteriorly :- Superior surface of 1st thoracic vertebra
On each side :- 1st rib

- Plane of inlet is directed downwards and forwards with an obliquity of 45°

Partition at inlet of Thorax :-

- Partition in 2 halves → Right, left — with cleft b/w.
- Each half is covered by fascia - Sibson fascia or Supraplevral membrane
- Membrane — Δ in shape
  - Apex ⟶ attached to tip of tranverse Process of 7th cervical vertebra
  - Base → inner border of first rib

- Sibson fascia is → flattened tendon of scalenus minimus

Structure Passing through the Inlet of Thorax :-

- Viscera - Trachea
  - Oesophagus
  - apics of lungs with pleura
  - thymus

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- Large vessels - Brachiocephalic artery on right side
  left common carotid artery → on left side
  left subclavian artery
  Right and left brachiocephalic vein
- Smaller vessels - Right and left internal thoracic artery
  Right and left superior intercastal arteries
  Right and left first posterior intercastal veins
  Inferior thyroid veins
- Nerves - Right & left phrenic nerve, vagus nerve, Sympathetic trunks
- Muscles - Sternohyoid
  Sternothyroid
  longus coli

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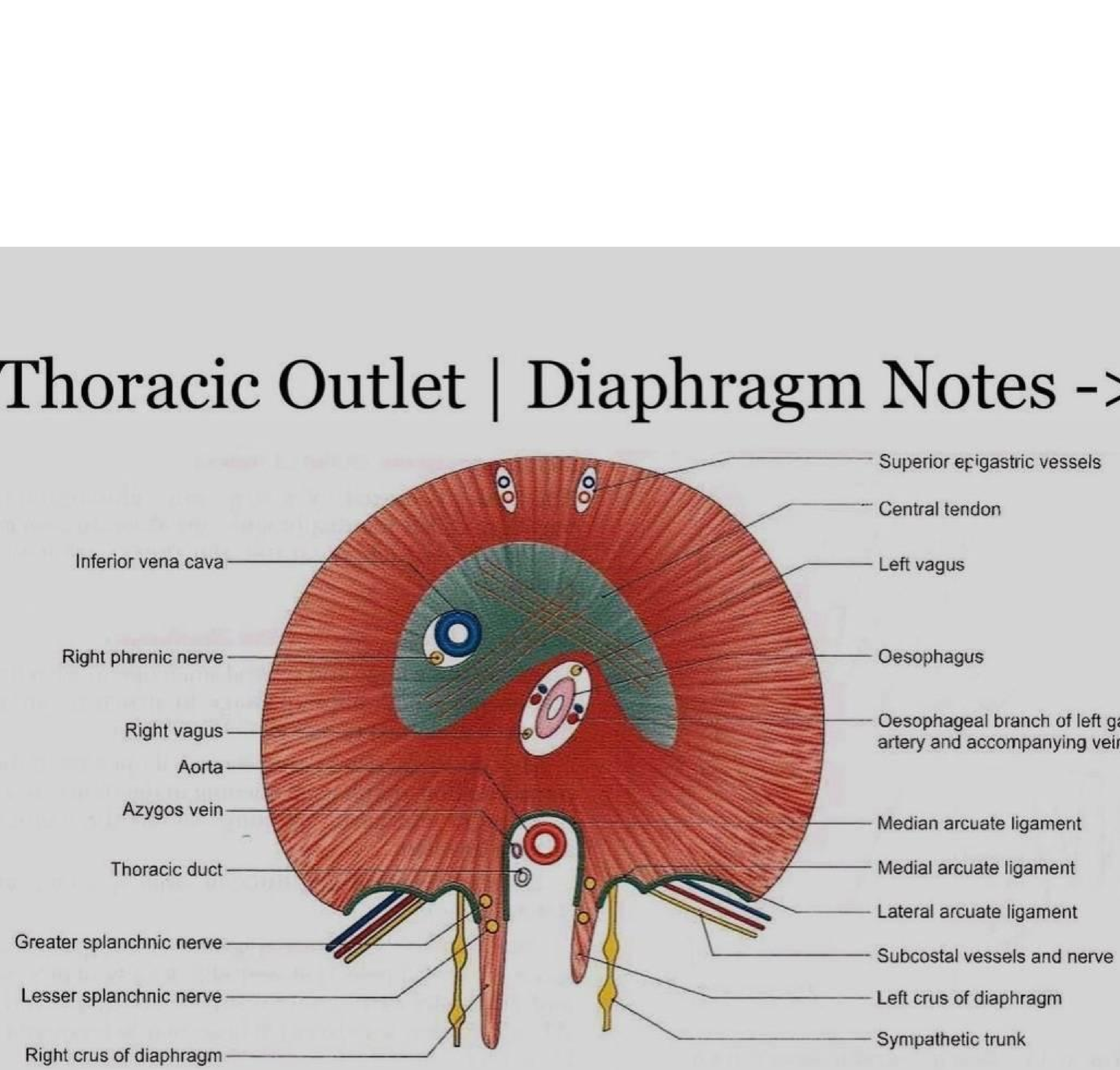


Fig. 12.16: Structures passing through the diaphragm

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Outlet of Thorax...

Introduction :— Outlet of thorax is broad end of thorax.
↳ which sourround upper part of abdominal cavity.
↳ By it seperate it by diaphragm.

Boundaries :—

Ant. — Infrasternal angle blw the 2 costal margins.
Post. — Inf. surface of the body of the 12th thoracic vertebrae.
On each side — coastal margin formed by the cartilage of 7th to 12th rib.

Diaphragm at outlet of thorax ...

Outlet of thorax is closed by large musculotendenious partition cld diaphragm.

Origin :—
Ant. — Xiphoid process.
Post. — L_1 to L_5 vertebrae.
Each side lower 6 costal cartilage.

Insertion :— Central tendon of diaphragm.

Innervation:- Phrenic nerve.

Structure passing throug thorax:—

o — **Large opening** — Vena cava opening $\xrightarrow{\text{structure passing}}$ IVC
Rt phrenic nerve
Lymphatic of liver.

Oesophageal opening — Oesophagus
Both vagal trunk.

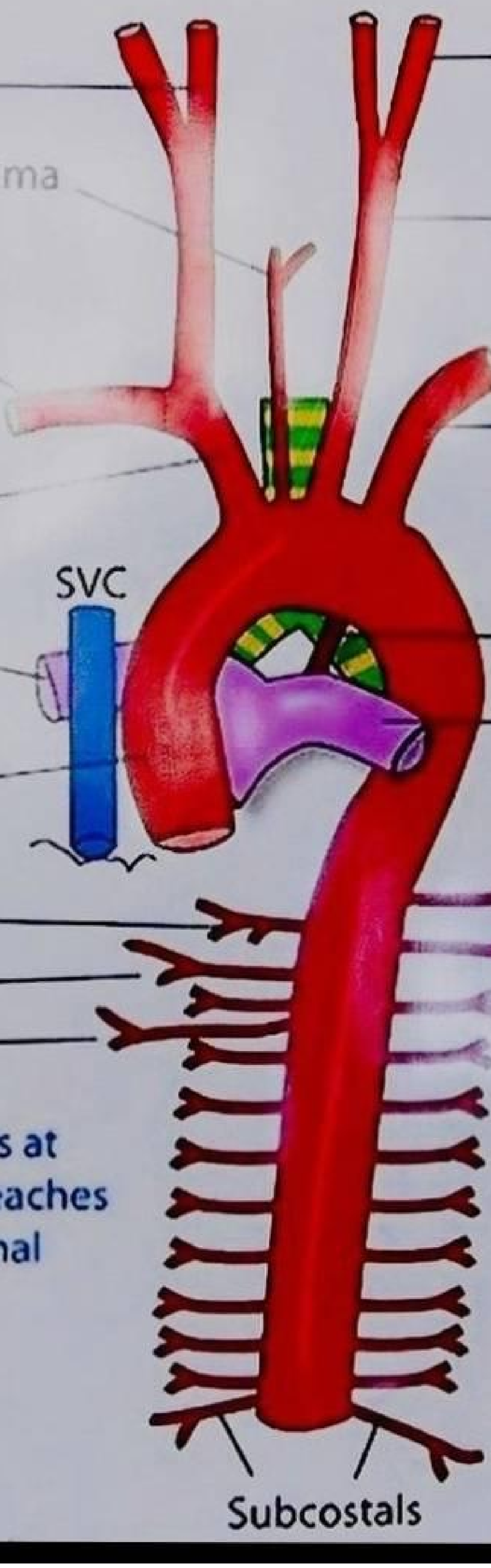
Aortic opening — Aorta
Thoracic duct
Azygos vein.

o — **Small opening** — Sup. epigastric vessel.
Sympathetic trunk
o — ~~Sub costal vessel~~ & nerve — passing behind lat. arcute ligament.
o — **Subcostal trunk** pass behind medial arcute lig.
o — ~~greater Least~~ greater & lesser ~~Splenic~~ Splanchnic nerve pairs each crus.

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THORACIC AORTA



Internal carotid

Arteria thyroidea ima

Right subclavian artery

Brachiocephalic artery

Right pulmonary artery

Ascending aorta

Oesophageal

Right bronchial

Mediastinal

Ascending aorta starts at aortic orifice (T4/5), reaches T3/4 & ends T4/5 sternal angle

SVC

Subcostals

External carotid

Common carotid artery

Left subclavian artery

Ligamentum arteriosum

Left pulmonary artery

Left bronchial

Posterior intercostals (3-11) (Spaces 1 & 2 from superior intercostal from costocervical trunk)

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Arch of Aorta ...

↳ Continuation of ascending aorta.
↳ Situated in mediastinum.

Course :-

↳ Begin behind the upper border of the sec. right sternochondral joint.
↳ Runs upward, backward & bifurcation of trachea.
↳ It thus arches over root of lung.
↳ Ends at lower border of body of 4th thoracic vertebrae.

Relation :-

Anteriorly & to the left.
- Lt phrenic nerve.
- Lower cervical cardiac branch of the Lt vagus.
- Lt vagus.
- Lt sup. intercostal vein.
- Lt pleura & lung.

Posteriorly & to the Right.
- Trachea.
- Oesophagus.
- Lt recurrent laryngeal nerve.
- Thoracic duct.

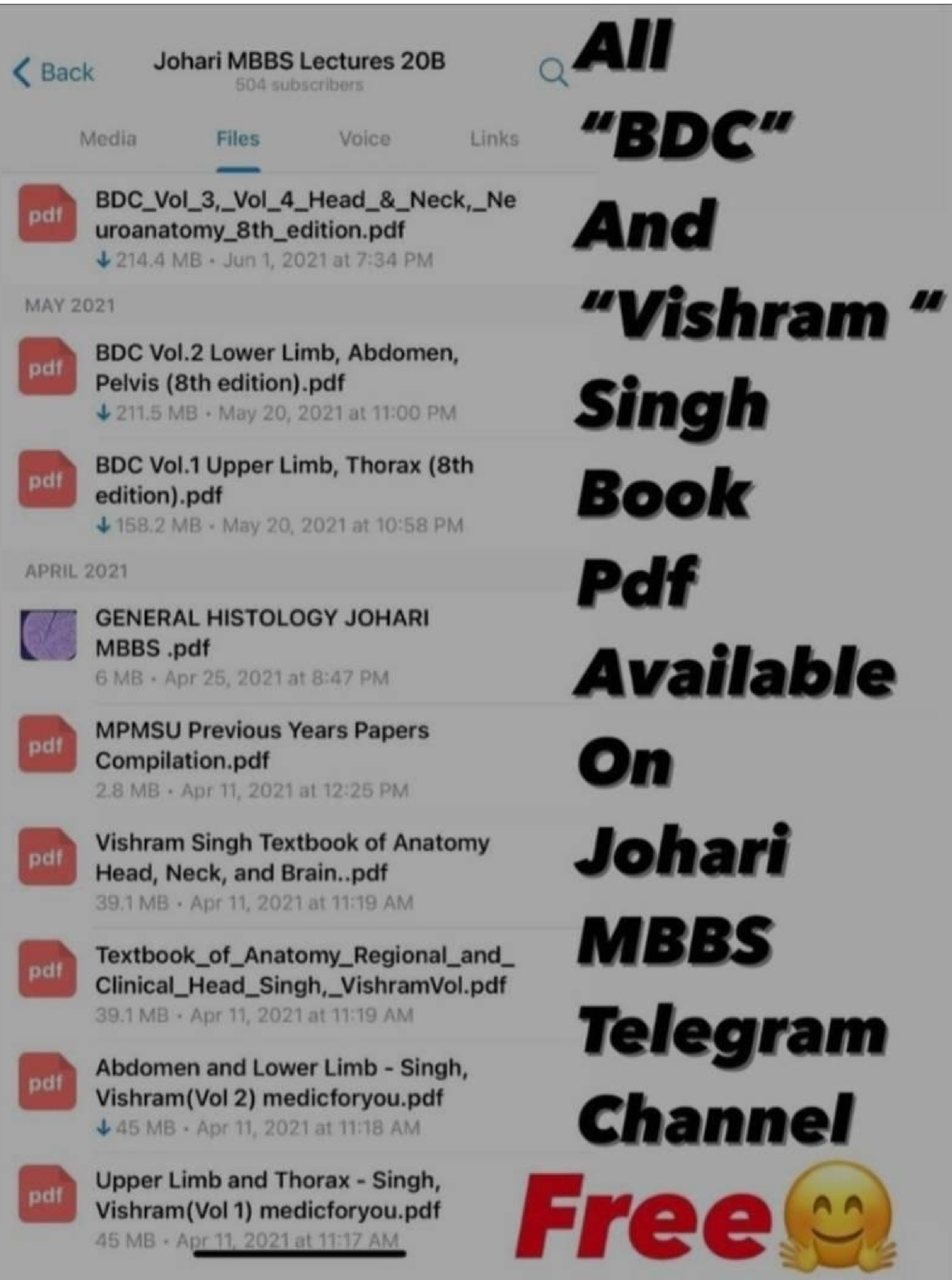
Superior -
3 branches of arch of aorta.
A) Brachiocephalic.
B] Lt common carotid.
C) Lt subclavian artery.

Inf. —
Bifurcation of the pulmonary trunk.
Lt Bronchus.

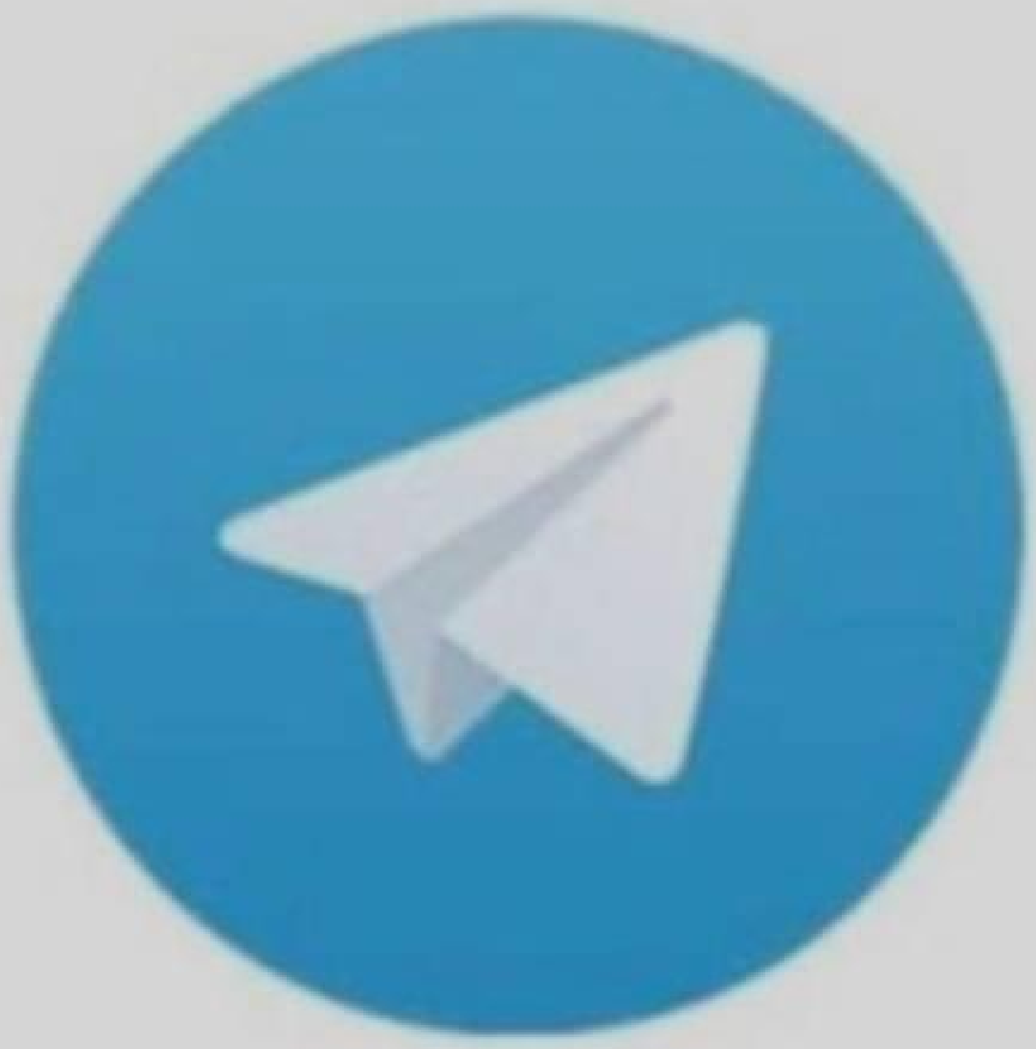
Branches :-
Brachiocephalic artery. → divides → Rt common carotid, Rt subclavian arteries.
Lt common carotid artery.
Lt subclavian artery.

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